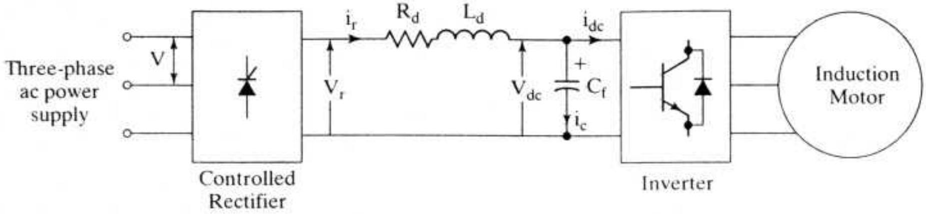


Figure 7.22 DC link: details of the motor drive

DC link: For a realistic dc link, as shown in Figure 7.22, the relationships are

$$v_{dc} = \frac{1}{C_f p}(i_r - i_{dc}) \tag{7.51}$$

$$v_{dc} > 0, i_r \geq 0 \tag{7.52}$$

with the constraint that the dc link voltage never become less than zero. The controlled rectifier's output voltage is given by

$$v_r = 1.35V_s \cos \alpha \tag{7.53}$$

which, in terms of the link parameters, is written as

$$v_r = v_{dc} + (R_d + pL_d)i_r \tag{7.54}$$

By neglecting the inverter and cable losses, the dc link power can be equated to the input power of the induction motor; it is given by

$$v_{dc}i_{dc} = \frac{3}{2}(v_{qs}^e v_{qs}^e + v_{ds}^e i_{ds}^e) \tag{7.55}$$

and the zero-sequence power, $v_o i_o$, is zero for a balanced three-phase system. By substituting for v_{qs}^e and v_{ds}^e from (7.44) and (7.45) into (7.55), the dc link current is defined as

$$i_{dc} = i_{qs}^e \cos\left(\theta_s + \frac{\pi}{3}\right) + i_{ds}^e \sin\left(\theta_s + \frac{\pi}{3}\right) \tag{7.56}$$

Stator-referred dc link: The dc link and stator q and d axes voltages are at different magnitudes. For uniformity of treatment, the dc link variables need to be referred to the stator of the induction motor. It is done by defining a fictitious dc link whose relationship to the actual dc link variables is as follows:

$$v'_{dc} = \frac{2}{3}v_{dc} \tag{7.57}$$

$$i'_{dc} = i_{dc} \tag{7.58}$$

and hence the impedance transformation ratio is

$$\frac{v'_{dc}}{i'_{dc}} = \frac{2}{3}\frac{v_{dc}}{i_{dc}}$$

i.e., expressed in terms of the impedances as

$$\frac{Z'_{dc}}{Z_{dc}} = \frac{2}{3} \tag{7.59}$$

where the fictitious and actual dc link impedances are defined, respectively, as follows:

$$Z'_{dc} = \frac{v'_{dc}}{i'_{dc}} \tag{7.60}$$

$$Z_{dc} = \frac{v_{dc}}{i_{dc}} \tag{7.61}$$

From the impedance-transformation ratio, the dc link filter constants can be written as

$$R'_d = \frac{2}{3} R_d \tag{7.62}$$

$$L'_d = \frac{2}{3} L_d \tag{7.63}$$

$$C'_f = \frac{3}{2} C_f \tag{7.64}$$

Similarly,

$$v'_r = \frac{2}{3} v_r \tag{7.65}$$

and the fictitious rectified dc voltage is

$$v'_r = v'_{dc} + (R'_d + pL'_d)i'_r \tag{7.66}$$

The power balance is expressed as,

$$v'_r i'_r = (v'_{dc} + (R'_d + pL'_d)i'_r)i'_r = v'_{dc}i'_r + R'_d(i'_r)^2 \tag{7.67}$$

and the fictitious input power to the inverter is

$$v'_{dc}i'_{dc} = \frac{2}{3} v_{dc}i_{dc} = \frac{2}{3}\left[\frac{3}{2}\{v^e_{qs}i^e_{qs} + v^e_{ds}i^e_{ds}\}\right] = v^e_{qs}i^e_{qs} + v^e_{ds}i^e_{ds} \tag{7.68}$$

This expression makes the dc link compatible to the motor model, in the sense that the sum of the q and d axis powers is equal to the dc link power. The d and q axis voltages then can be written in terms of the fictitious variables as

$$_s = \frac{2}{3} v_{dc} \cos\left(\theta_s + \frac{\pi}{3}\right) = \frac{2}{3}\frac{3}{2} v'_{dc} \cos\left(\theta_s + \frac{\pi}{3}\right) = \{v'_r - (R'_d + L'_d p)i'_r\}\cos\left(\theta_s + \frac{\pi}{3}\right) \tag{7.69}$$

Similarly, the d axis voltage is derived as

$$v^e_{ds} = \frac{2}{3} v'_r \sin\left(\theta_s + \frac{\pi}{3}\right) - (R'_d + L'_d p)i'_r \sin\left(\theta_s + \frac{\pi}{3}\right) \tag{7.70}$$

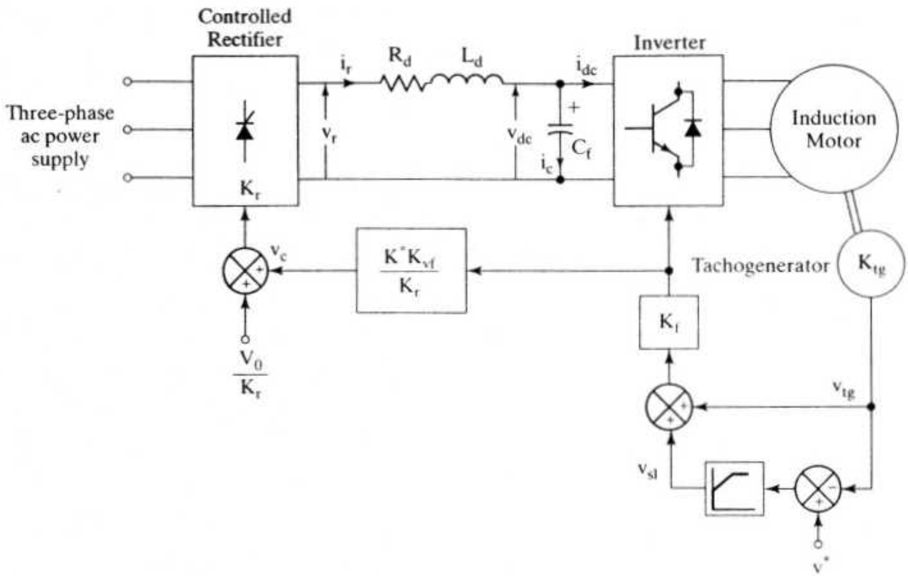


Figure 7.23 Control-block diagram of the volts/Hz-controlled induction motor drive

Substituting equations (7.69) and (7.70) into equation (7.38) yields a set of equations for simulating the dynamics of the volts/Hz-controlled induction motor drive.

Controller: The drive diagram, with its blocks, is shown in Figure 7.23. The external input v^* is

$$-V_{cm} < v^* < +V_{cm}, \text{ V} \tag{7.71}$$

where V_{cm} is the maximum control voltage and is either ± 10 or ± 5 V. v^* is proportional to the commanded speed of the motor, and the proportionality constant K^* is defined as

$$K^* = \frac{v^*}{\omega_r^*} = \frac{V_{cm}}{\max\{\omega_r^*\}}, \text{ volt/(rad/sec)} \tag{7.72}$$

where ω_r^* is the commanded electrical rotor speed in rad/sec. The tachogenerator in the associated gain block is adjusted to have its maximum output corresponding to $\pm V_{cm}$ for control compatibility. Hence the gain of the tachogenerator and the filter is

$$K_{tg} = \frac{V_{cm}}{\omega_m} = \frac{V_{cm}}{\omega_r/(P/2)} = \frac{P}{2} \cdot K^* \tag{7.73}$$

The maximum slip speed corresponds to the maximum torque of the induction motor; to avoid statically unstable and low-efficiency operation, the motor is not allowed to exceed the maximum slip speed, whose value, therefore, is

$$\max\{v_{sl}\} = K^* \omega_{sl(max)} \text{ (V)} \tag{7.74}$$

The sum of the slip-speed signal and the rotor electrical-speed signal corresponds to the synchronous speed; hence, the gain of the frequency transfer block is

$$K_f = \frac{1}{2\pi K^*} \text{(Hz/volt)} \tag{7.75}$$

The stator frequency and dc link voltage are written as

$$f_s = K_f\left(K^*\omega_{sl} + \frac{P}{2}K^*\omega_m\right) = K_f K^*(\omega_r + \omega_{sl}), \text{Hz} \tag{7.76}$$

The control voltage of the output rectifier, obtained from a previous derivation, is

$$v_c = \frac{2.22}{K_r}(V_o + K_{vf}f_s)$$

Hence, the output of the rectifier is

$$v_r = 2.22K_r\{V_o + K_f K^* K_{vf}(\omega_r + \omega_{sl})\} \tag{7.77}$$

where K_r is the gain of the controlled rectifier, and the offset voltage and slip speed are generalized as follows:

$$V_o = I_{sr}R_s \tag{7.78}$$

and

$$\omega_{sl} = f_n\{v^* - v_{tg}\} = f_n\{v^* - \omega_m K_{tg}\} = f_n\{K^*\omega_r^* - K^*\omega_r\} \tag{7.79}$$

where I_{sr} is the rated stator current and f_n is the speed controller function.

The controller usually is a PI controller in tandem with a phase lead–lag network.

The synchronous speed, then, is given in terms of the controller constants and variables as

$$\omega_s = 2\pi f_s = 2\pi K_f K^*(\omega_r + \omega_{sl}) \tag{7.80}$$

By using equations (7.77) to (7.80), (7.69), and (7.70) in the equation (7.38), the modeling of the inverter-fed induction motor drive is obtained. Solving these equations by using fourth-order Runge–Kutta gives the time responses of the rectifier, inverter, and motor variables. They can be effectively employed to study the dynamic interaction of the drive system and to design a suitable controller and compensators.

Simulation Results: The drive given in Example 7.1 is simulated for a one-p.u.-step speed command from standstill. All the important variables of the induction motor drive are shown in Figure 7.24 in parts (i) and (ii). Notable is the torque behavior in this drive. Torque is unable to follow its command transiently, because the rotor flux linkages oscillate, resulting in large stator current and torque transients. In an actual drive system this could be avoided by limiting the stator currents. The controlled rectifier currents reach peaks of 14 p.u. This could be controlled by increasing the filter inductance or by reducing the rate of change of

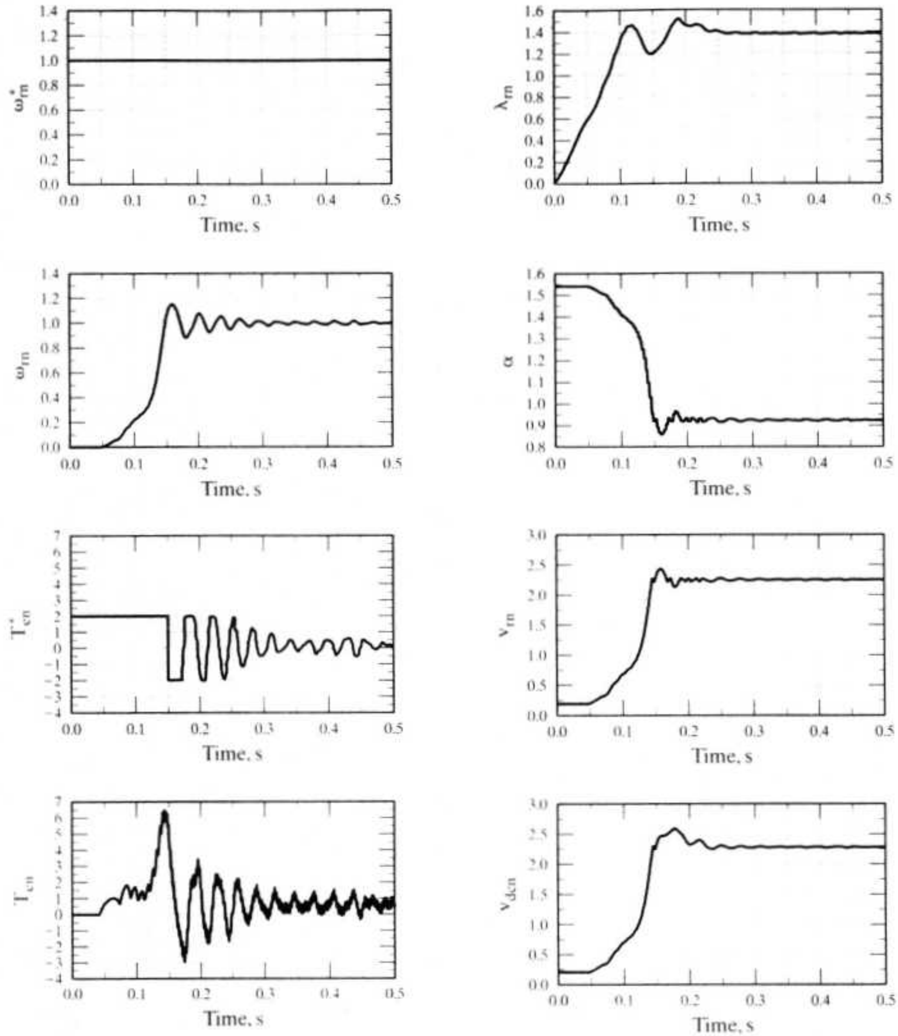


Figure 7.24(i) Dynamic performance of the volts/Hz-controlled induction motor drive system—Part (i)

the dc link voltage. The ac input line current reaches a maximum of 5.5 p.u., and the current in the capacitor has a maximum of 9 p.u.. On the inverter side, the stator current has a maximum value of 5 p.u.. Although this is smaller than the controlled rectifier currents, it is not acceptable; it increases the inverter rating to multiple times that of the induction motor. In practice, current limiting is set in the controller to keep this under control.

7.4.5.5 Small-signal responses. The small-signal responses are important in evaluating the stability and bandwidth of the motor drive. They are assessed by

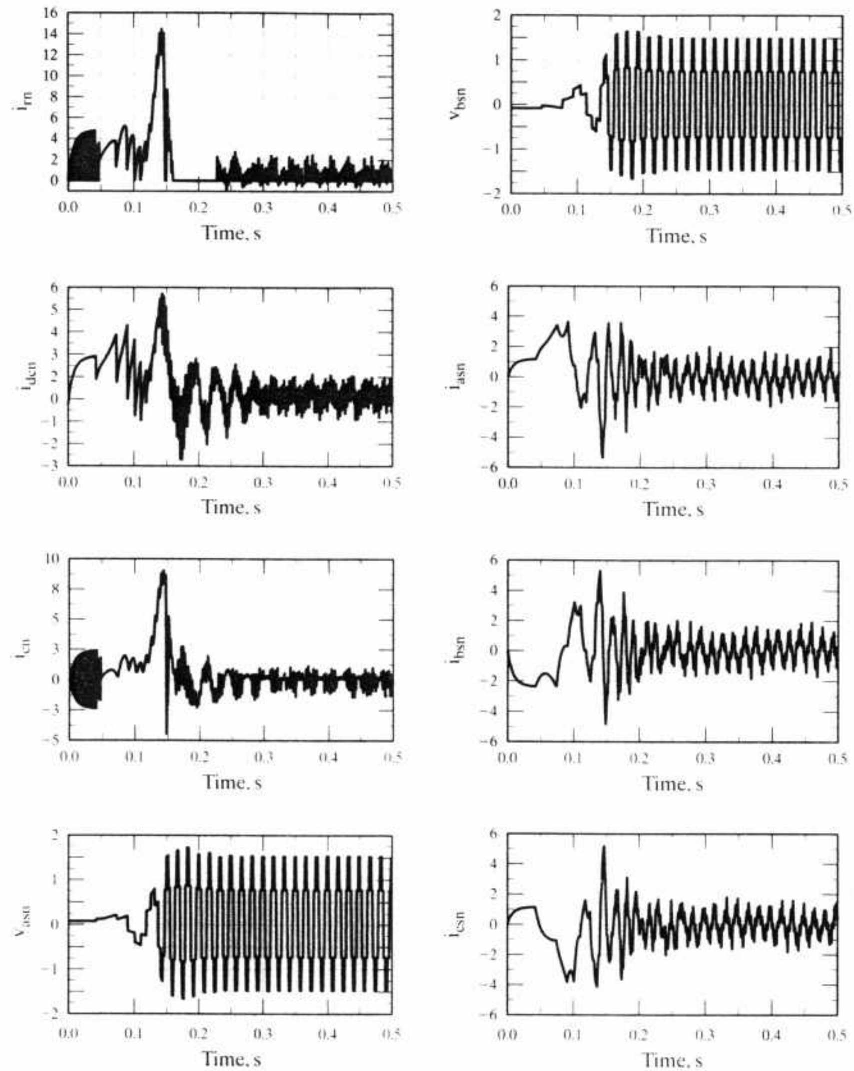


Figure 7.24(ii) Dynamic performance of the volts/Hz-controlled induction motor drive system—Part (ii)

considering only the fundamental of the input voltages, contrary to the dynamic simulation. Consideration of fundamentals only reduces the inputs and outputs to dc quantities; hence, a perturbation around the steady-state operating point results in small-signal equations.

The steady-state operating point is determined by neglecting the harmonics in the input line voltages given in equations (7.11) to (7.13). The d and q voltages are calculated, by using the transformation given in equation (7.41), from the phase voltages, which, in turn, are derived from line voltages. It is worth mentioning at this juncture that choosing the phase voltages in the following form will result in simple *d* and *q* voltages.

$$v_{as} = \frac{2}{\pi} V_{dc} \cos \omega_s t \tag{7.81}$$

$$v_{bs} = \frac{2}{\pi} V_{dc} \cos\left(\omega_s t - \frac{2\pi}{3}\right) \tag{7.82}$$

$$v_{cs} = \frac{2}{\pi} V_{dc} \cos\left(\omega_s t + \frac{2\pi}{3}\right) \tag{7.83}$$

resulting in

$$v_{qs}^e = \frac{2}{\pi} V_{dc} \tag{7.84}$$

$$v_{ds}^e = 0$$

Denoting the steady-state values by an additional subscript '0', the steady-state currents are evaluated from equation (7.71) by making $p = 0$ and by maintaining ω_r constant. The equations then become algebraic; hence, the currents and subsequently the torque can be evaluated. The steady-state currents, neglecting the dc-link filter dynamics, are

$$\begin{bmatrix} i_{qso}^e \\ i_{dso}^e \\ i_{qro}^e \\ i_{dro}^e \end{bmatrix} = \begin{bmatrix} R_s & \omega_s L_s & 0 & \omega_s L_m \\ -\omega_s L_s & R_s & -\omega_s L_m & 0 \\ 0 & (\omega_s - \omega_r)L_m & R_r & (\omega_s - \omega_r)L_r \\ -(\omega_s - \omega_r)L_m & 0 & -(\omega_s - \omega_r)L_r & R_r \end{bmatrix}^{-1} \begin{bmatrix} v_{qso}^e \\ v_{dso}^e \\ 0 \\ 0 \end{bmatrix} \tag{7.85}$$

$$T_e = \frac{3}{2}\frac{P}{2} L_m(i_{qso}^e i_{dro}^e - i_{dso}^e i_{qro}^e)$$

Neglecting the dc-link filter dynamics, we perturb the motor equations around a steady-state operating point, and small-signal equations are derived similar to those in Chapter 5, section 5.9. Likewise, the loop equations describing the relationship between the speed command and voltage command and stator frequency are perturbed to obtain the small-signal equations. Combining the motor and feedback-loop small-signal equations gives the system equations. They can be used to study various transfer functions, stability, and small-signal transient responses and can be used for purposes of speed-controller design.

7.4.5.6 Direct steady-state evaluation. The steady state has been calculated so far by using the fundamental of the input voltages only. The steady-state performance for the actual input voltages, including harmonics, is necessary to select the rating of the converter–inverter switches and for computation of losses and derating of the induction motor. The steady state then is calculated either by using steady-state-harmonic equivalent circuits and summing the responses, or directly, by matching the boundary conditions. The harmonic equivalent-circuit approach has the conceptual advantage of simplicity but carries the disadvantage that its accuracy is limited by the number of harmonics considered in the input voltages. The direct steady-state evaluation overcomes this disadvantage but used to be limited by the requirement of a computer for solution. This method is derived and discussed in the section that follows.

The direct method exploits the symmetry of input voltages and currents in the steady state. They are symmetric over a given interval, in this case 60°, so their boundaries are matched to extract an elegant solution. Considering the input voltages, it has been proven in section 7.4.5.4 that they are periodic for every 60 electrical degrees. Hence, for a linear system (the case whenever speed is constant in the induction motor), the response must also be periodic. That is, the stator and rotor currents are periodic. The derivations are given below.

The induction motor equations in synchronously rotating reference frame are written in state-variable form as

$$\dot{X}_1 = A_1X_1 + B_1U_1 \tag{7.86}$$

where

$$X_1 = [i^e_{qs} \quad i^e_{ds} \quad i^e_{qr} \quad i^e_{dr}]^t \tag{7.87}$$

$$A_1 = Q^{-1}P_1 \tag{7.88}$$

$$B_1 = Q^{-1} \tag{7.89}$$

$$u_1 = [v^e_{qs} \quad v^e_{ds} \quad 0 \quad 0]^t \tag{7.90}$$

$$Q = \begin{bmatrix} L_s & 0 & L_m & 0 \\ 0 & L_s & 0 & L_m \\ L_m & 0 & L_r & 0 \\ 0 & L_m & 0 & L_r \end{bmatrix} \tag{7.91}$$

$$P_1 = \begin{bmatrix} -R_s & -\omega_s L_s & 0 & -\omega_s L_m \\ \omega_s L_s & -R_s & \omega_s L_m & 0 \\ 0 & -(\omega_s - \omega_r)L_m & -R_r & -(\omega_s - \omega_r)L_r \\ (\omega_s - \omega_r)L_m & 0 & (\omega_s - \omega_r)L_r & -R_r \end{bmatrix} \tag{7.92}$$

From equations (7.44) and (7.45), the voltages are written in state-space form as

$$\begin{bmatrix} pv^e_{qs} \\ pv^e_{ds} \end{bmatrix} = \begin{bmatrix} 0 & -\omega_s \\ \omega_s & 0 \end{bmatrix} \begin{bmatrix} v^e_{qs} \\ v^e_{ds} \end{bmatrix} \tag{7.93}$$

Equations (7.86) and (7.87) are combined to give

$$\begin{bmatrix} \dot{X}_1 \\ \dot{X}_2 \end{bmatrix} = \begin{bmatrix} A_1 & B_1 \\ 0 & S \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \tag{7.94}$$

where

$$X_2 = [v^e_{qs} \quad v^e_{ds}]^t \tag{7.95}$$

$$S = \begin{bmatrix} 0 & -\omega_s \\ \omega_s & 0 \end{bmatrix} \tag{7.96}$$

and matrices A_1 and B_1 are of compatible dimensions. The equation (7.94), in a compact form, is expressed as

$$\dot{X} = AX \tag{7.97}$$

where

$$X = [X_1 \quad X_2]^t \tag{7.98}$$

$$A = \begin{bmatrix} A_1 & B_1 \\ 0 & S \end{bmatrix} \tag{7.99}$$

The solution of equation (7.97) is written as,

$$X(t) = e^{At}X(0) \tag{7.100}$$

where the initial steady-state vector $X(0)$ is to be evaluated to compute $X(t)$ and the electromagnetic torque. It is found by the fact that the state vector has periodic symmetry; hence,

$$X\left(\frac{\pi}{3\omega_s}\right) = S_1X(0) \tag{7.101}$$

where S_1 is evaluated later.

The boundary condition for the currents is

$$X_1\left(\frac{\pi}{3\omega_s}\right) = X_1(0) \tag{7.102}$$

The boundary-matching condition for the voltage vector is obtained by expanding (7.49) and (7.50) and substituting (7.44) and (7.45) into them. The direct axis voltage is

$$v_{qsII}^e(\theta_s) = \frac{2}{3}V_{dc}\cos\left(\theta_s + \frac{\pi}{3} - \frac{\pi}{3}\right) = \frac{2}{3}V_{dc}\left\{\cos\left(\theta_s + \frac{\pi}{3}\right)\cos\frac{\pi}{3} + \sin\left(\theta_s + \frac{\pi}{3}\right)\sin\frac{\pi}{3}\right\}$$

$$= \frac{1}{2}v_{qsI}^e(\theta_s) + \frac{\sqrt{3}}{2}v_{dsI}^e(\theta_s) \tag{7.103}$$

Similarly,

$$v_{dsII}^e = \frac{1}{2}v_{dsI}^e - \frac{\sqrt{3}}{2}v_{qsI}^e \tag{7.104}$$

Hence,

$$X_2\left(\frac{\pi}{3\omega_s}\right) = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} X_2(0) = S_2X_2(0) \tag{7.105}$$

where

$$S_2 = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \tag{7.106}$$

S_1 is obtained from the equations (7.102) and (7.105) as

$$S_1 = \begin{bmatrix} I & 0 \\ 0 & S_2 \end{bmatrix} \tag{7.107}$$

where I is a 4×4 identity matrix.

Substituting equation (7.101) into (7.100), we get

$$X\left(\frac{\pi}{3\omega_s}\right) = S_1X(0) = e^{A\left(\frac{\pi}{3\omega_s}\right)}X(0) \tag{7.108}$$

Hence,

$$\left[S_1 - e^{A\left(\frac{\pi}{3\omega_s}\right)}\right]X(0) = 0 \tag{7.109}$$

or

$$WX(0) = 0 \tag{7.110}$$

where

$$W = \left[S_1 - e^{A\left(\frac{\pi}{3\omega_s}\right)}\right] = \begin{bmatrix} W_1 & W_2 \\ W_3 & W_4 \end{bmatrix} \tag{7.111}$$

where W_1 is 4×4, W_2 is 2×4, W_3 is 2×4 and W_4 is 2×2. It can be proven that W_3 is a null matrix. Expanding only the upper row in equation (7.110) gives the following relationship:

$$W_1X_1(0) + W_2X_2(0) = 0 \tag{7.112}$$

from which the steady-state current vector $X_1(0)$ is obtained as

$$X_1(0) = -W_1^{-1}W_2X_2(0) \tag{7.113}$$

Having evaluated the initial current vector, we could use it in equation (7.100) to evaluate currents for one full cycle and the electromagnetic torque. A sample set of steady-state waveforms is shown in Figures 7.25(i) and (ii). The peaking of the current at no-load and then its becoming quasi-sinusoidal for full load is significant to determine the peak rating of the devices in the inverter. Note that this approach is suitable for steady-state calculation of six-stepped voltage inputs, irrespective of the control strategy used. The electromagnetic torque has a sixth-harmonic ripple in the six-stepped-voltage-fed induction motor drive that deserves scrutiny. Its cause and effects are considered in later sections.

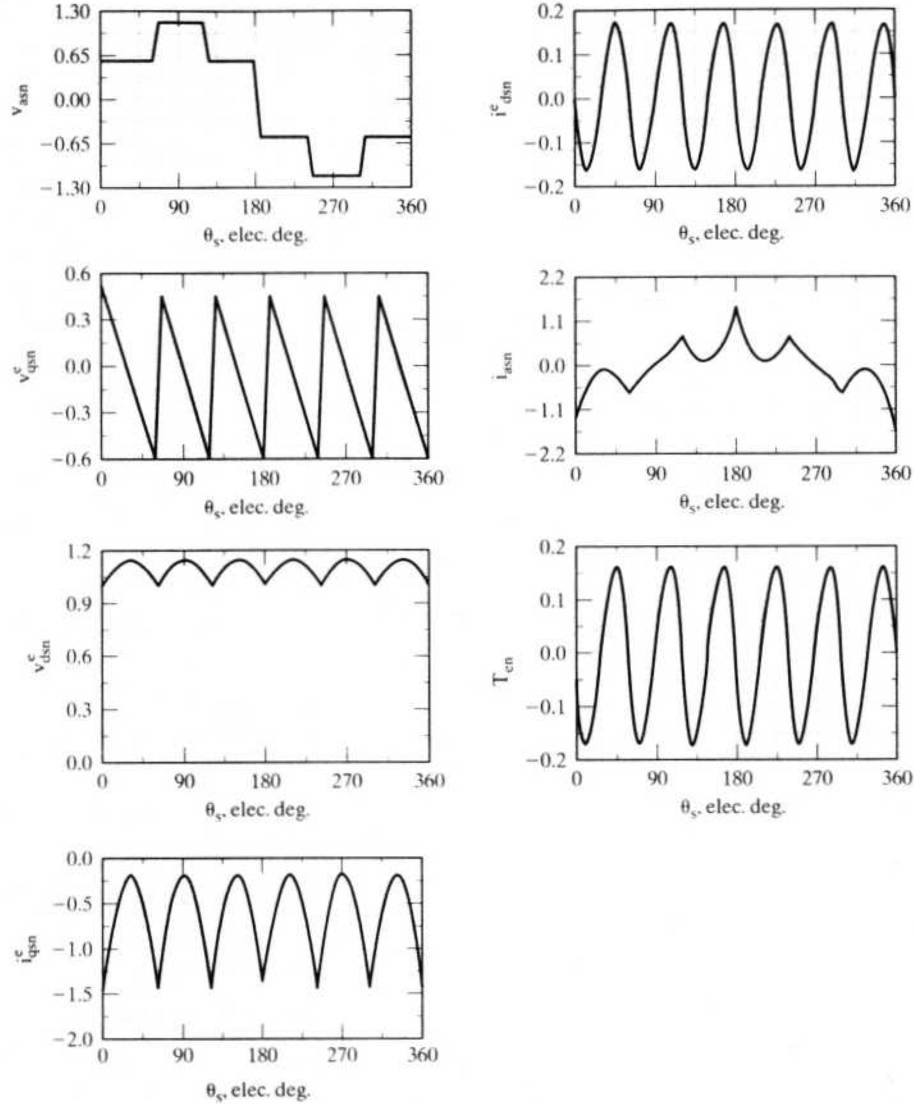


Figure 7.25(i) Steady-state waveforms of the VSIM drive with six-step stator voltages at no-load in p.u.

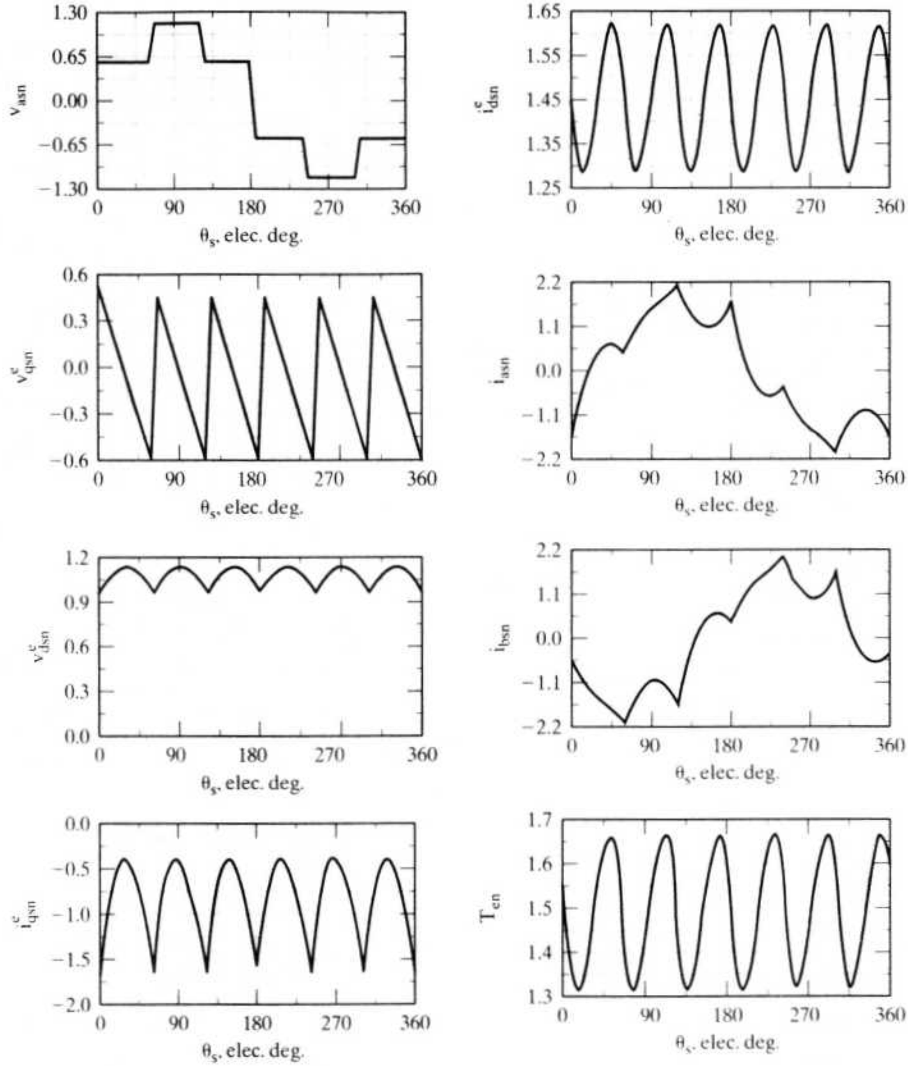


Figure 7.25(ii) Steady-state waveforms of the VSIM drive with six-step stator voltages under load in p.u.

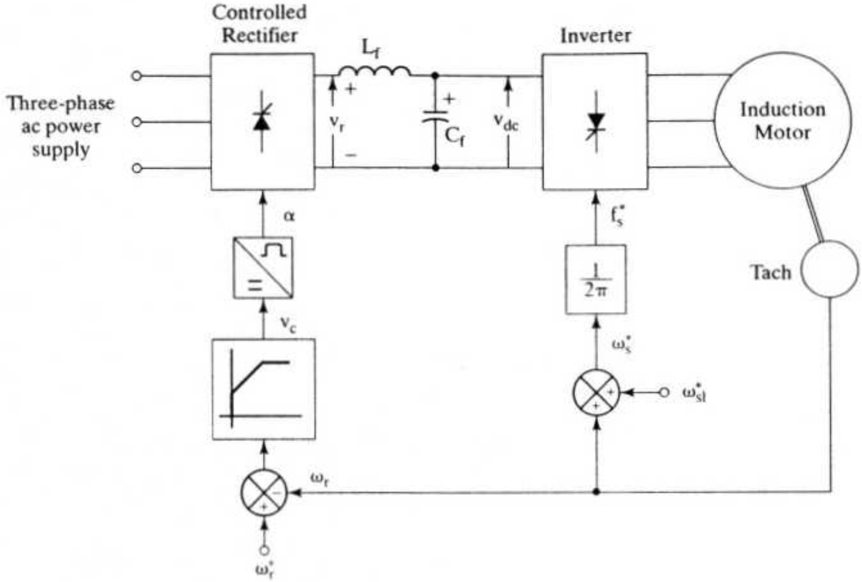


Figure 7.26 Constant-slip-speed drive strategy

7.4.6 Constant Slip-Speed Control

7.4.6.1 Drive strategy. The slip speed of the induction motor is maintained constant; hence, for various rotor speeds, the slip will be varying, as is seen from the following expressions:

$$\omega_s = \omega_r + \omega_{sl} \tag{7.114}$$

$$\omega_{sl} = s\omega_s = \text{constant} \tag{7.115}$$

from which the slip is obtained as

$$s = \frac{\omega_{sl}}{\omega_s} = \frac{\omega_{sl}}{\omega_r + \omega_{sl}} \tag{7.116}$$

The varying slip control places the drive operation on the static torque–speed characteristics. To maintain the slip speed constant, it is necessary to know the rotor speed, so this scheme involves rotor-speed estimation or measurement for feedback control. The stator frequency is obtained by summing the slip speed and the electrical rotor speed. The required input voltages to the induction motor are made to be a function of the speed-error signal, as is shown in Figure 7.26. In place of a proportional controller, a PI controller eliminates the steady-state error in the rotor speed. Note that a negative speed error clamps the bus voltage at zero, and triggering angles of greater than 90° are not allowed. Given the configuration of the drive, it cannot regenerate: the stator electrical speed is always maintained greater than the

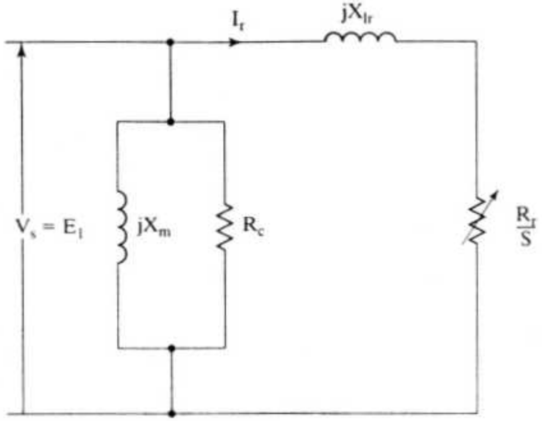


Figure 7.27 Simplified equivalent circuit considered for the steady-state analysis of the slip-controlled induction motor drive

rotor speed. Hence, this drive is restricted to one-quadrant operation only. Alternate implementation of dc link voltage control is effected by implementing a volts/Hz controller, using f_s^*. Note that this alternate scheme will not give a closed-loop speed control.

7.4.6.2 Steady-state analysis. The characteristics of the constant-slip-speed induction motor are derived in this section from the simplified equivalent circuit of the induction motor. Only the fundamental of the applied voltages is considered at this stage. Considering the rotor and magnetic circuit shows that the equivalent circuit amounts to the one shown in Figure 7.27. The slip speed is a constant; hence, in terms of the slip speed, the rotor current is derived as

$$I_r = \frac{E_1}{\left(\frac{R_r}{s} + jX_{lr}\right)} = \frac{E_1/\omega_s}{\left(\frac{R_r}{\omega_{sl}} + jL_{lr}\right)} \tag{7.117}$$

and the electromagnetic torque is

$$T_e = \frac{P}{2} \cdot \frac{P_a}{\omega_s} = 3 \cdot \frac{P}{2} \cdot \frac{I_r^2 R_r}{s\omega_s} = 3 \cdot \frac{P}{2} \cdot \frac{I_r^2 R_r}{\omega_{sl}} \tag{7.118}$$

Substituting for I_r from equation (7.117) into equation (7.118) yields

$$T_e = 3\frac{P}{2} \cdot \frac{E_1^2}{\omega_s^2} \cdot \frac{\left(\frac{R_r}{\omega_{sl}}\right)}{\left(\frac{R_r}{\omega_{sl}}\right)^2 + (L_{lr})^2} \tag{7.119}$$

By rearranging all the constants into one term, we get

$$T_e = K_{tv}\left(\frac{E_1^2}{\omega_s^2}\right) \tag{7.120}$$

where the torque constant for this strategy is defined as

$$K_{tv} = \frac{3\frac{P}{2}\left(\frac{R_r}{\omega_{sl}}\right)}{\left(\frac{R_r}{\omega_{sl}}\right)^2 + (L_{lr})^2} \tag{7.121}$$

Neglecting stator impedance amounts to making the air gap emf equal to the applied stator voltage; hence, the torque is given by

$$T_e \cong K_{tv}\left(\frac{V_s}{\omega_s}\right)^2 \tag{7.122}$$

where V_s is the stator voltage per phase.

Note that the torque is independent of rotor speed, implying its capability to produce a torque even at zero speed. This feature is essential in many applications where a starting or holding torque needs to be produced, such as in robotics.

A diagram of torque vs. stator phase voltage is shown in Figure 7.28 for three slip speeds. Here, the stator impedance has not been neglected, and its consideration has reduced the torque, notably at low speeds. The motor data has been taken from Example 7.1.

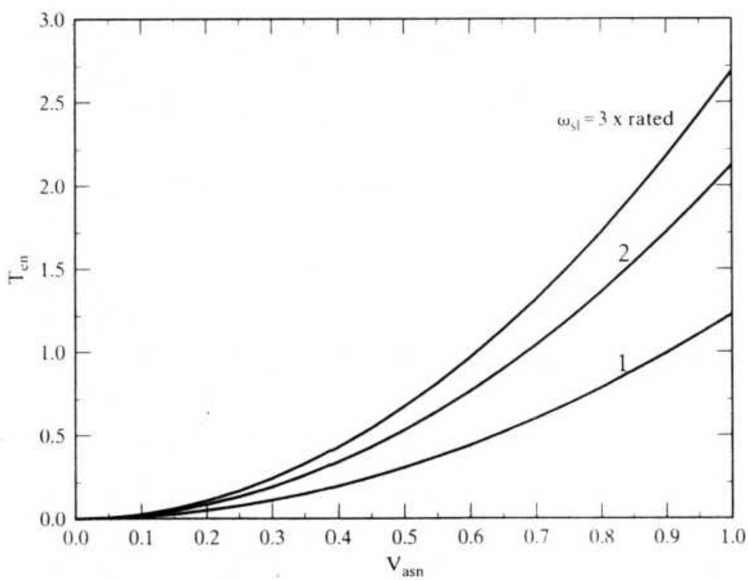


Figure 7.28 Torque vs. applied voltage for various slip speeds at rated stator frequency in p.u.

Example 7.2

Draw the performance characteristics of the constant-slip-speed-controlled induction motor drive given in Example 7.1. The slip speed is maintained at 9 rad/sec. The converter combination is a controlled rectifier and a six-step voltage source inverter. The drive uses a volts/Hz strategy for controlling the dc link voltage.

Solution The steps involved are as follows:

$$\omega_r = \omega_m \times \frac{P}{2}$$

$$\omega_s = \omega_r + \omega_{sl}$$

$$K_{vf} = \frac{V_{s(rated)}}{f_s} = \frac{V_{s(rated)}}{(\omega_s/2\pi)}$$

where

$$V_s = 0.45V_{dc} = K_{vf}f_s$$

By using the fundamental equivalent circuit, the stator, magnetizing, and rotor currents are computed. From the currents, the motor power factor, torque, output, and efficiency are evaluated. The performance characteristics are shown in Figure 7.29.

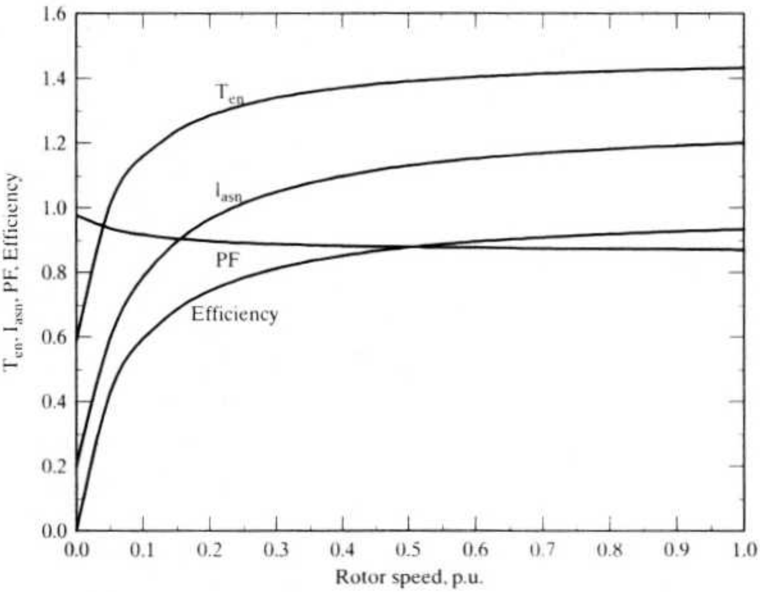


Figure 7.29 Performance characteristics of slip-controlled induction motor drive at ω_{sl} = 9 rad/sec in p.u.

7.4.7 Constant Air Gap Flux Control

7.4.7.1 Principle of operation. Constant air gap flux control resolves the induction motor into an equivalent separately-excited dc motor in terms of its speed of response but not in terms of decoupling of the flux and torque channels. Having constant air gap flux linkages amounts to

$$\lambda_m = L_m i_m = \frac{E_1}{\omega_s} \tag{7.123}$$

which, substituted into equation (7.119), yields the electromagnetic torque as

$$T_e = 3\frac{P}{2}\cdot\lambda_m^2\cdot\frac{\left(\frac{R_r}{\omega_{sl}}\right)}{\left(\frac{R_r}{\omega_{sl}}\right)^2 + (L_{lr})^2} \tag{7.124}$$

Assuming the air gap flux linkage is maintained constant, the torque is

$$T_e = K_{tm}\cdot\frac{\left(\frac{R_r}{\omega_{sl}}\right)}{\left(\frac{R_r}{\omega_{sl}}\right)^2 + (L_{lr})^2} \tag{7.125}$$

where the torque constant in the control strategy is written as

$$K_{tm} = 3\frac{P}{2}\cdot\lambda_m^2 \tag{7.126}$$

Now the electromagnetic torque is dependent only on the slip speed, as is seen from equation (7.125). Such a feature signifies a very important phenomenon, in that the slip speed can be varied instantly, making the torque response instantaneous. A fast torque response paves the way for a high-performance motor drive, suitable for demanding applications, thus replacing the separately-excited dc motor drives. The above facts are true only if the air gap flux is maintained constant. That task is compounded by, for example, the saturation of the machine or the need for a sensor to measure the air gap flux. Even if air gap flux is regulated accurately, the torque is not a linear function of the slip speed. Hence, for a torque drive, the slip speed has to be programmed to generate linear characteristics between torque and its commanded value. Note that this programming of the slip speed has to account for the sensitivity of the rotor resistance and leakage inductance. In particular, the rotor resistance will usually have a wide variation (from 0.8 to 2 times its nominal value at ambient temperature) and this complicates the control task.

7.4.7.2 Drive strategy. The fact that the constant air gap flux requires control of the magnetizing current necessitates stator current control of the induction motor, apart from its stator frequency. A voltage-source inverter drive with inner current loops would transform it into a variable-current, variable-frequency source.

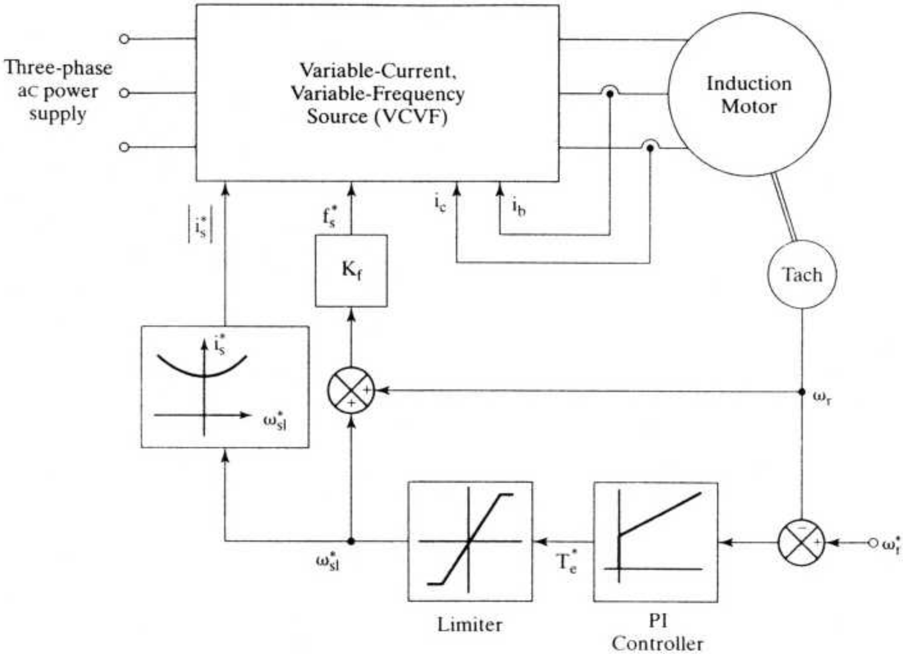


Figure 7.30 Drive strategy for constant-air gap-flux-controlled induction motor drive

More of this is to come in later sections. Assuming that such a current-regulated variable-frequency source is available, the drive strategy is shown in Figure 7.30.

The slip-speed command is generated from the speed-control error, which in turn is added to the rotor speed signal to provide the frequency-command signal. The same slip-speed command determines the stator current magnitude, maintaining the air gap flux constant. This aspect is shown as a function generator block. For reasons of saturation and sensitivity of rotor constants, it is preferable to have on-line computation of the stator current from the slip-speed command. The current command is appropriately translated into three-phase stator-current commands. Stator-current feedback control ensures that the stator-current commands are enforced both in magnitude and in phase.

Example 7.3

Draw the steady-state performance characteristics of a constant-air gap-flux-controlled induction motor drive. The motor data are given in Example 7.1.

Solution The various steps involved in the steady-state performance computation are as follows:

Step 1: Calculate the rated value of air gap flux from the equivalent circuit including stator impedence when the motor has a rated electromagnetic torque. Calculate magnetizing current and hence E_1

Step 2: For a given stator frequency, calculate the air gap emf.

Step 3: Starting with a slip, and from the air gap emf, calculate the rotor current, and then find the stator current, power factor, torque, applied stator voltage, and efficiency. The equations are

$$I_r = \frac{E_1}{\left(\frac{R_r}{s} + jX_{lr}\right)} \tag{7.127}$$

$$I_s = I_r + I_m \tag{7.128}$$

$$V_{as} = E_1 + I_s(R_s + j\omega_s L_{ls}) \tag{7.129}$$

$$T_e = 3I_r^2 \cdot \frac{R_r}{s\omega_s} \cdot \frac{P}{2} \tag{7.130}$$

$$\text{Power factor} = \cos\phi = \frac{\text{Real}(V_{as}I_s^*)}{|V_{as}I_{as}|} \tag{7.131}$$

$$\begin{aligned} \text{Efficiency} &= \frac{\text{Mechanical power output}}{\text{Electrical power input}}, s > 0 \\ &= \frac{\text{Electrical power output}}{\text{Mechanical power input}}, s < 0 \\ &= 0 \qquad , s = 0 \end{aligned} \tag{7.132}$$

Step 4: Increment the slip, and go to step 3.

Step 5: Increment the stator frequency, and go to step 2.

Step 6: Draw torque vs. speed for various stator frequencies and rotor and stator currents and voltage and torque vs. speed for rated stator frequency.

The normalized electromagnetic torque vs. speed for various stator frequencies is shown in Figure 7.31. The torque characteristic is uniquely symmetric for both the motoring and generating mode and for every stator frequency. The asymmetry in the torque characteristics for the motoring and generating regions in the volts/Hz-controlled drive is due to the lack of control of air gap flux linkages. The stator current and voltage vs. speed are shown in Figure 7.32 for rated stator frequency.

Constant air gap flux linkages result in higher demand for stator phase voltages, as is seen in Figure 7.32. This is due to the increasing rotor and hence stator currents and the consequent increase in voltage drops across the stator impedances. For torques less than 2 p.u., the increase in the stator voltage, although very small, could exceed the dc bus supply. This necessitates proper selection of the induction-motor voltage rating for an available dc supply. High-transient-torque demands of four to five p.u. in emerging applications such as electric vehicle propulsion need to consider the stator voltage demand at higher speeds.

Example 7.4

Derive stator current magnitude in terms of the motor parameters, slip speed, and magnetizing current to implement a constant-air gap-flux-linkages drive system.

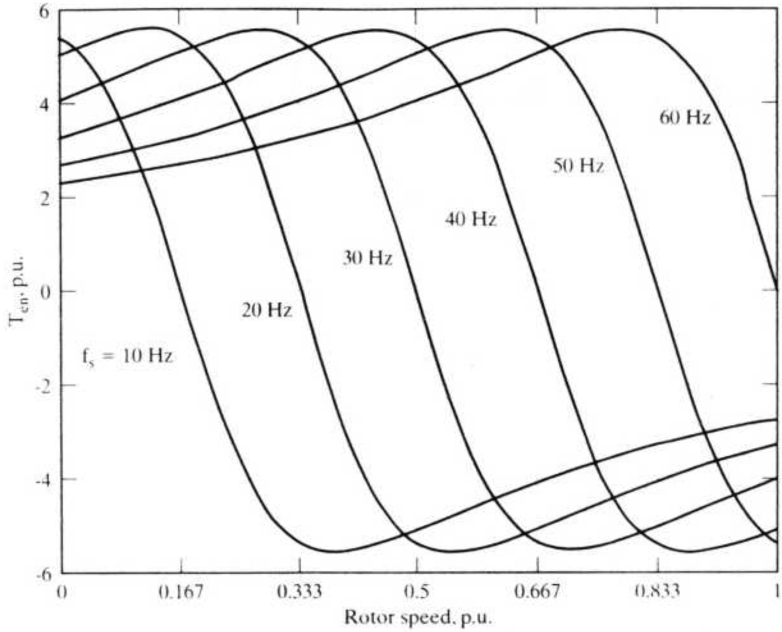


Figure 7.31 Normalized torque vs. speed characteristics of the constant-air gap-flux-controlled induction motor drive for different stator frequencies

Solution The equivalent circuit shown in Figure 7.27 is used to derive the stator current magnitude, as follows:

The rotor current is

$$I_r = \frac{E_1}{\left(\frac{R_r}{s} + jX_{lr}\right)} = \frac{jX_m I_m}{\left(\frac{R_r}{s} + jX_{lr}\right)} = \frac{jL_m I_m s\omega_s}{R_r + js\omega_s L_{lr}} = \frac{jL_m I_m \omega_{sl}}{R_r + j\omega_{sl} L_{lr}}$$

$$I_s = I_r + I_m = I_m\left[1 + \frac{jL_m\omega_{sl}}{R_r + j\omega_{sl}L_{lr}}\right] = I_m\left[\frac{R_r + j\omega_{sl}(L_{lr} + L_m)}{R_r + j\omega_{sl}L_{lr}}\right]$$

$$|I_s| = I_m\sqrt{\frac{R_r^2 + (\omega_{sl}L_r)^2}{R_r^2 + (\omega_{sl}L_{lr})^2}}$$

The stator current is dependent upon the rotor self-inductance and leakage inductance and on the resistance of the rotor apart from the slip-speed and magnetizing current. Because the flux linkages are controlled, the variation of the rotor self-inductance and leakage inductance will not vary significantly, but the rotor resistance due to the temperature and slip frequency variations will significantly vary, and, in that case, the stator current has to be made a function of the rotor resistance to maintain the flux constant. It is an important consideration in the design of the controller in this drive scheme.

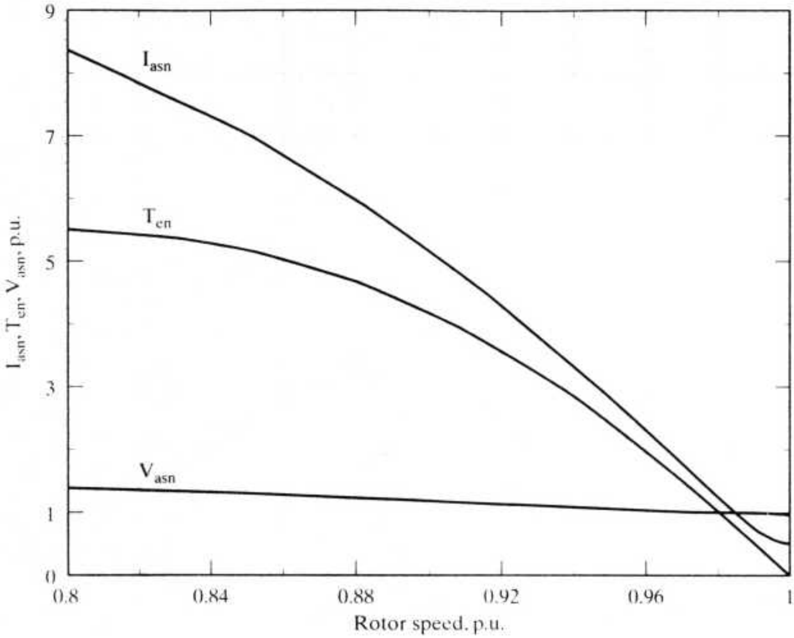


Figure 7.32 Torque, voltage, and current vs. speed at rated stator frequency

7.4.8 Torque Pulsations

7.4.8.1 General. Six-stepped-voltage waveforms are rich in harmonics. These time harmonics produce respective rotor current harmonics, which in turn interact with the fundamental air gap flux, generating harmonic torque pulsations. The torque pulsations are undesirable: they generate audible noise, speed pulsations, and losses, thus decreasing the thermal capability of the motor and eventually derating the motor. Even though the magnitude of the torque pulsations can be evaluated from the steady-state computation by using boundary-matching conditions, it is simple to calculate the magnitude of each harmonic torque pulsation by using the harmonic equivalent circuit of the induction motor. This method has the advantage of singling out the dominant torque pulsation; when its source is identified, methods can be devised to control it.

7.4.8.2 Calculation of torque pulsations. By using the Fourier series of the line voltages given in equations (7.11) to (7.13), the fundamental, fifth, and seventh harmonics of the phase voltages are derived as

$$V_{as1} = \frac{2}{\pi} V_{dc} \sin(\omega_s t - 30^\circ) \tag{7.133}$$

$$V_{as5} = \frac{2}{5\pi} V_{dc} \sin(-5\omega_s t - 30^\circ) \tag{7.134}$$

$$V_{as7} = \frac{2}{7\pi} V_{dc} \sin(7\omega_s t - 150°) \tag{7.135}$$

The fifth harmonic is rotating opposite to the fundamental, whereas the seventh is in the same direction as the fundamental. Therefore, the air gap flux linkages due to the fifth- and seventh-harmonic currents field, are revolving at six times the synchronous speed relative to the fundamental air gap flux.

The sixth-harmonic torque pulsation is created by

(i) the fundamental air gap flux linkages, interacting with the fifth- and seventh-harmonic rotor currents;

(ii) the fundamental rotor current, interacting with the fifth- and seventh-harmonic air gap flux linkages.

The various variables are computed, as follows, from the two equivalent-circuit diagrams shown in Figure 7.33.

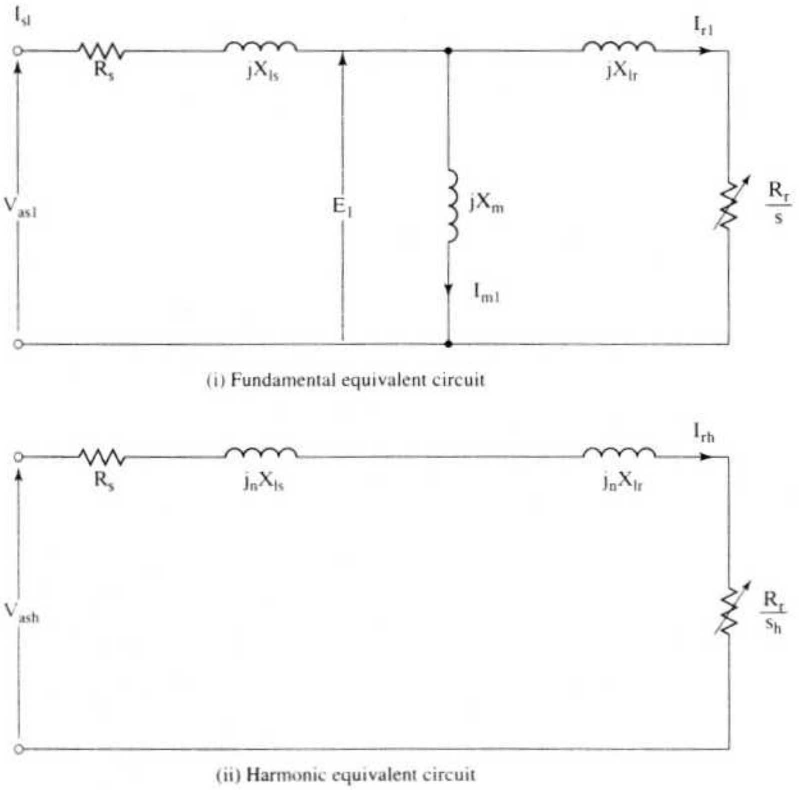


Figure 7.33 Equivalent circuit of the induction motor

The harmonic slip for a harmonic of order h, derived in Chapter 6, is given as

$$s_h \cong \frac{h \pm 1}{h}, \begin{cases} + \text{ for h odd} \\ - \text{ for h even} \end{cases} \tag{7.136}$$

Hence,

$$\begin{aligned} s_5 &\cong \frac{6}{5} \\ s_7 &\cong \frac{6}{7} \end{aligned} \tag{7.137}$$

The fundamental, fifth-, and seventh-harmonic mutual flux linkages are

$$\lambda_{m1} = L_m I_{m1} \tag{7.138}$$

$$\lambda_{m5} = L_m I_{m5} = \frac{I_{r5}}{5\omega_s}\left(\frac{R_r}{s_5} + j5X_{lr}\right) \tag{7.139}$$

$$\lambda_{m7} = \frac{I_{r7}}{7\omega_s}\left(\frac{R_r}{s_7} + j7X_{lr}\right) \tag{7.140}$$

and the harmonic rotor currents are given by

$$I_{r5} = \frac{V_{as5}}{\left(R_s + \dfrac{R_r}{s_5}\right) + j5(X_{ls} + X_{lr})} \tag{7.141}$$

$$I_{r7} = \frac{V_{as7}}{\left(R_s + \dfrac{R_r}{s_7}\right) + 7(jX_{ls} + X_{lr})} \tag{7.142}$$

but at frequencies above 0.3 p.u., the rotor peak currents can be approximated as

$$I_{r5} \cong \frac{V_{as5}}{5(X_{ls} + X_{lr})} \cong \frac{2V_{dc}}{5\pi}\left(\frac{1}{5X_{eq}}\right) \cong \left(\frac{2V_{dc}}{\pi}\right)\left(\frac{1}{25X_{eq}}\right) \tag{7.143}$$

where the equivalent leakage reactance is given as

$$X_{eq} = (X_{ls} + X_{lr}) \tag{7.144}$$

and

$$I_{r7} \cong \left(\frac{2V_{dc}}{\pi}\right)\left(\frac{1}{49X_{eq}}\right) \tag{7.145}$$

By substituting these approximated values of rotor currents into equations (7.139) and (7.140) and neglecting the resistances in comparison to the harmonic leakage reactances, the harmonic peak flux linkages are found to be

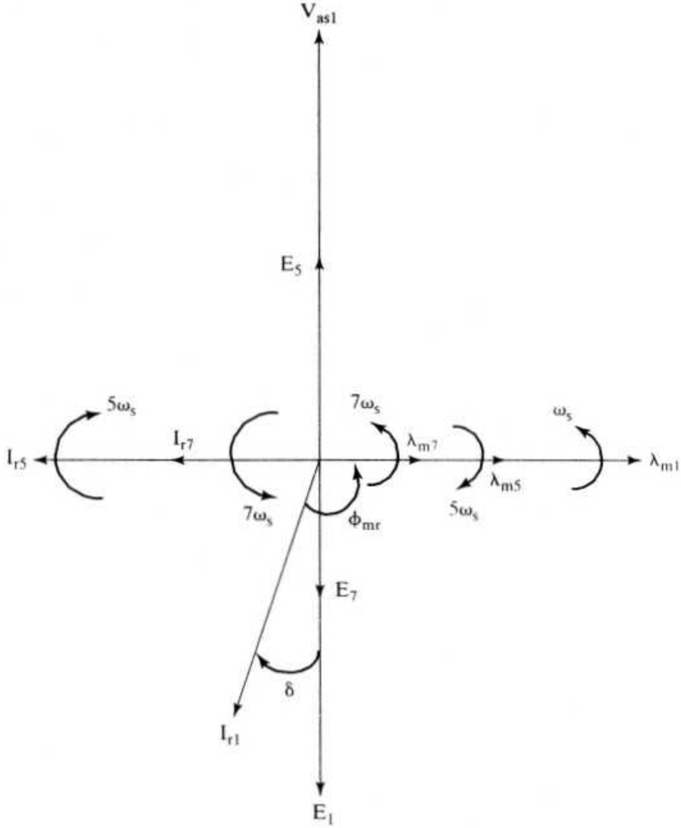


Figure 7.34 Phasor diagram of the induction motor, with harmonic components

$$\lambda_{m5} \cong I_{r5}L_{1r} \cong \frac{2V_{dc}}{25\pi\omega_s}\left(\frac{L_{1r}}{L_{eq}}\right) \tag{7.146}$$

$$\lambda_{m7} \cong \frac{2V_{dc}}{49\pi\omega_s}\left(\frac{L_{1r}}{L_{eq}}\right) \tag{7.147}$$

where

$$L_{eq} = \frac{X_{eq}}{\omega_s} \tag{7.148}$$

These variables are represented in a phasor diagram, as shown in Figure 7.34. The fundamental torque is then computed as

$$T_{e1} = 3\frac{P}{2}\lambda_{m1}I_{r1}\sin\phi_{mr} \tag{7.149}$$

and the peak of the fundamental mutual flux linkages is

$$\lambda_{m1} \cong \frac{V_{as1}}{\omega_s} = \frac{2V_{dc}}{\pi\omega_s} \tag{7.150}$$

Substituting equation (7.150) into equations (7.146) and (7.147) yields the harmonic flux linkages in terms of fundamental flux linkages:

$$\lambda_{m5} = \frac{\lambda_{m1}}{25}\left(\frac{L_{lr}}{L_{eq}}\right) \tag{7.151}$$

$$\lambda_{m7} = \frac{\lambda_{m1}}{49}\left(\frac{L_{lr}}{L_{eq}}\right) \tag{7.152}$$

The sixth-harmonic torque in the anticlockwise direction is

$$\begin{aligned} T_{e6} = \frac{3}{2}\frac{P}{2}\,[\lambda_{m1}(I_{r7} - I_{r5})\sin 6\omega_s t + I_{r1}\{\lambda_{m7}\sin(6\omega_s t + 90^\circ + \delta) \\ + \lambda_{m5}\sin(-6\omega_s t + 90^\circ + \delta)\}] \end{aligned} \tag{7.153}$$

where

$$90^\circ + \delta = \phi_{mr} \tag{7.154}$$

δ is very small in practice; assuming it is zero, the sixth-harmonic torque is approximated as

$$T_{e6} \cong \frac{3}{2}\frac{P}{2}\,[\lambda_{m1}(I_{r7} - I_{r5})\sin 6\omega_s t + I_{r1}(\lambda_{m7} + \lambda_{m5})\cos 6\omega_s t] \tag{7.155}$$

The torque is divided by an additional factor of 2 because the flux linkage and rotor currents are peak values. Normalizing the sixth-harmonic torque in terms of its fundamental torque yields

$$\begin{aligned} T_{e6n} &= \frac{T_{e6}}{T_{e1}} = \frac{(I_{r7} - I_{r5})}{I_{r1}\sin\phi_{mr}}\sin 6\omega_s t + \left(\frac{L_{lr}}{L_{eq}}\right)\frac{\left(\frac{1}{25} + \frac{1}{49}\right)}{\sin\phi_{mr}}\cos 6\omega_s t \\ &\cong \frac{(I_{r7} - I_{r5})}{I_{r1}}\sin 6\omega_s t + (0.0604)\left(\frac{L_{lr}}{L_{eq}}\right)\cos 6\omega_s t \end{aligned} \tag{7.156}$$

Similarly, twelfth-harmonic pulsating torque can be evaluated from the eleventh- and thirteenth-harmonic input voltages. In most of the applications, the dominant-pulsating-torque calculation is sufficient.

These pulsating torques are smoothed by the rotor and load inertia at high speeds, but, at low speeds, they might induce speed ripples, which are undesirable. They could excite and resonate the critical frequencies of the motor drive. In such cases, corrective measures have to be taken. Elimination of undesirable pulsating torques is discussed subsequently.

Example 7.5

Calculate the sixth-harmonic pulsating torque for the motor drive given in Example 7.1, neglecting stator impedance. The operating points are at 60 Hz with 0 and 1 p.u. slip speeds.

Solution

Case (i) $\omega_{sl} = 0, s = 0 \therefore I_{r1} = 0$

There is slip speed in regard to the time-harmonic input voltages, which in turn generates rotor harmonic current and pulsating torque. The rms values are calculated; accordingly, the torque expression has the term 3 and not, as in the previous derivation, 3/2. The rotor harmonic currents and air gap flux linkages are

$$\lambda_{m1} = \frac{E_1}{\omega_s} \cong \frac{V_{as1}}{\omega_s} \cong \frac{200/\sqrt{3}}{2\pi \times 60} \cong 0.306 \text{ Wb} - \text{turn}$$

$$\lambda_{m5} \cong I_{r5} L_{1r}$$

$$\lambda_{m7} \cong I_{r7} L_{1r}$$

where

$$I_{r5} \cong \frac{V_{as5}}{5X_{eq}} = \frac{\left(\frac{V_{as1}}{5}\right)}{5X_{eq}} = \frac{200/\sqrt{3}}{25 \times 2\pi \times 60 \times 0.0037} = 3.31 \text{ A}$$

$$I_{r7} = \frac{V_{as1}}{49X_{eq}} = \frac{200/\sqrt{3}}{49 \times 1.395} = 1.69 \text{ A}$$

Hence,

$$\lambda_{m5} = 3.31 \times 0.0022 = 0.0073 \text{ Wb} - \text{turn}$$

$$\lambda_{m7} = 1.69 \times 0.0022 = 0.0038 \text{ Wb} - \text{turn}$$

$$T_{e6} = 3\frac{P}{2}[\lambda_{m1}(I_{r7} - I_{r5}) \sin 6\omega_s t + I_{r1}(\lambda_{m7} + \lambda_{m5}) \cos 6\omega_s t]$$

$$= 3 \times \frac{4}{2}\{(0.306)(1.69 - 3.31) \sin 6\omega_s t\} = -2.97 \sin 6\omega_s t$$

Peak absolute value of $T_{e6} = 2.97 \text{ N}\cdot\text{m} = \frac{2.97}{20.0} = 0.149$ p.u.

Case (ii) $\omega_{sl} = 6.882$ rad / sec

$$\lambda_{m1} = 0.306 \text{ Wb} - \text{turn}$$

and

$$s = \frac{\omega_{sl}}{\omega_s} = \frac{6.882}{377} = 0.0183$$

$$\therefore I_{r1} \cong \frac{V_{as1}}{\frac{R_r}{s} + jX_{lr}} = \frac{115.5}{\frac{0.183}{0.0183} + j0.837} = \frac{115.5}{10 + j0.837} = 11.51 \text{ A}$$

The rotor harmonic currents and harmonic air gap flux linkages are the same as in the case where slip is equal to zero. Hence, the sixth-harmonic pulsating torque is computed as

$$T_{e6} = 3 \times \frac{4}{2}\{(0.306)(1.69 - 3.31)\sin 6\omega_s t + 11.51(0.0073 + 0.0038)\cos 6\omega_s t\}$$

$$= -2.97 \sin 6\omega_s t + 0.766 \cos 6\omega_s t$$

$$\text{Peak value of } T_{e6} = \sqrt{2.97^2 + 0.766^2} = 3.06\ \text{N}\cdot\text{m} = \frac{3.06}{20} = 0.153 \text{ p.u.}$$

Note that these calculations neglect the stator impedance drop, because, at rated frequency, the effect of the stator impedence drop is small compared to that of the induced emf in the air gap. At low operating frequencies, the stator impedance drop will dominate, and hence reduce the induced emf and hence the torque pulsations too. The torque pulsations will fatigue the shaft and eventually lead to failure. This type of motor drive is unsuitable for position applications, because of high torque ripples.

7.4.8.3 Effects of time harmonics. The harmonic voltages produce harmonic currents that, in turn, generate not only torque pulsation but also increased losses in the form of copper and core losses. The net effect of the torque pulsation on the resultant average torque is zero, but the harmonic losses add to the heating of the machine. Since the induction machines are usually designed for sinusoidal inputs, the additional losses due to the harmonics tax the thermal capability of the motor.

For a given cooling arrangement of the motor, the allowable temperature rise for safe operation is prescribed. To maintain the motor within its class of operation, the motor then needs to be derated. It is quite unavoidable with converter-controlled motors, but the magnitude of derating can be kept to a minimum by controlling the harmonic contents of the voltage and current inputs to the induction motor.

The induction motor has the following losses:

(i) stator copper loss;
(ii) rotor copper loss;
(iii) core loss;
(iv) friction and windage loss;
(v) stray load loss.

The presence of harmonics causes the stator and rotor copper losses to increase. As for the core loss, its increase is due to higher flux density caused by the harmonic components. Note that the fundamental and harmonic magnetizing currents increase the air gap mmf, and hence there is an increase in the flux density over the fundamental flux density. Compared to the harmonic copper losses, the increase in core loss is usually negligible for frequencies up to 100 Hz. For very high-speed machines operating at hundreds of Hz, the increase in core loss is considerable.

The friction and windage losses are independent of the harmonics. The stray losses are dependent on harmonics as some of stray-loss components are directly influenced by them. Its components are

(i) rotor zigzag loss,
(ii) stator end loss,
(iii) rotor end loss, and
(iv) other undefined losses.

Even though these losses constitute only a minor portion of the total losses, they are considerably increased by the harmonics. Their dependence on the harmonics is quantified and made available in references.

For standard machines, the total losses of inverter-fed machines can increase nearly as much as 50% compared to the losses resulting from a sinusoidal source. While this should be taken as a guideline, it is necessary to evaluate the losses for individual cases to appraise their suitability for specific applications.

Example 7.6

Consider the motor drive given in Example 7.1. Calculate the increase in stator and rotor copper losses and compare them with sinusoidal-source-produced losses at rated slip speed. Neglect any other loss and harmonics greater than 19.

Solution

$$I_{m1} = \frac{\lambda_{m1}}{L_m} = \frac{0.306}{0.05383} = 5.68 \text{ A}$$

$$I_{r1} = \frac{V_{as1}}{\frac{R_r}{s} + jX_{lr}} \cong \frac{115.5}{10} = 11.55 \text{ A}$$

$$I_{s1} \cong 11.55 + j5.68 = 12.87 \text{ A}$$

The stator and rotor resistive losses due to the fundamentals alone are

$$3I_{s1}^2 R_s = 3 \times (12.87)^2 \times 0.277 = 137.7 \text{ W}$$
$$3I_{r1}^2 R_s = 3 \times (11.55)^2 \times 0.183 = 73.2 \text{ W}$$

$$I_{rh} \cong \frac{V_{as1}}{h^2 X_{eq}}$$

$$I_{r5} = 3.31 \text{ A}$$

$$I_{r7} = 1.69 \text{ A}$$

$$I_{r11} = \frac{V_{as11}}{11 \times X_{eq}} = \frac{115.5/11}{11 \times 1.395} = 0.68 \text{ A}$$

$$I_{r13} = \frac{115.5/13}{13 \times 1.395} = 0.49 \text{ A}$$

$$I_{r17} = \frac{115.5/17}{17 \times 1.395} = 0.29 \text{ A}$$

$$I_{r19} = \frac{115.5/19}{19 \times 1.395} = 0.23 \text{ A}$$

$$I_{har} = \sqrt{I_{r5}^2 + I_{r7}^2 + I_{r11}^2 + I_{r13}^2 + I_{r17}^2 + I_{r19}^2}$$

$$= \sqrt{3.31^2 + 1.69^2 + .68^2 + 0.49^2 + 0.29^2 + 0.23^2} = 3.83 \text{ A}$$

$$\text{Stator current, } I_s = \sqrt{I_{s1}^2 + I_{har}^2} = \sqrt{12.87^2 + 3.83^2} = 13.43 \text{ A}$$

$$3I_s^2R_s = 3 \times 13.43^2 \times 0.277 = 149.82 \text{ W}$$

$$I_r \cong \sqrt{I_{r1}^2 + I_{har}^2} = \sqrt{11.55^2 + 3.83^2} = 12.16 \text{ A}$$

The stator and rotor resistive losses, including harmonics from 5th to 19th, are

$$3I_r^2R_r = 3 \times 12.16^2 \times 0.183 = 81.29 \text{ W}$$

$$\text{Increase in stator and rotor copper losses} = (149.82 - 137.9) + (81.29 - 73.2) = 20.2 \text{ W}$$

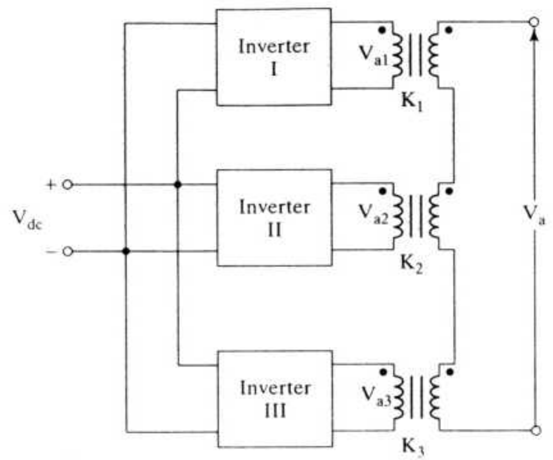
$$\text{\% increase over sinusoidal losses} = \frac{20.2}{137.7 + 73.2} \times 100 = 9.57\%$$

7.4.9 Control of Harmonics

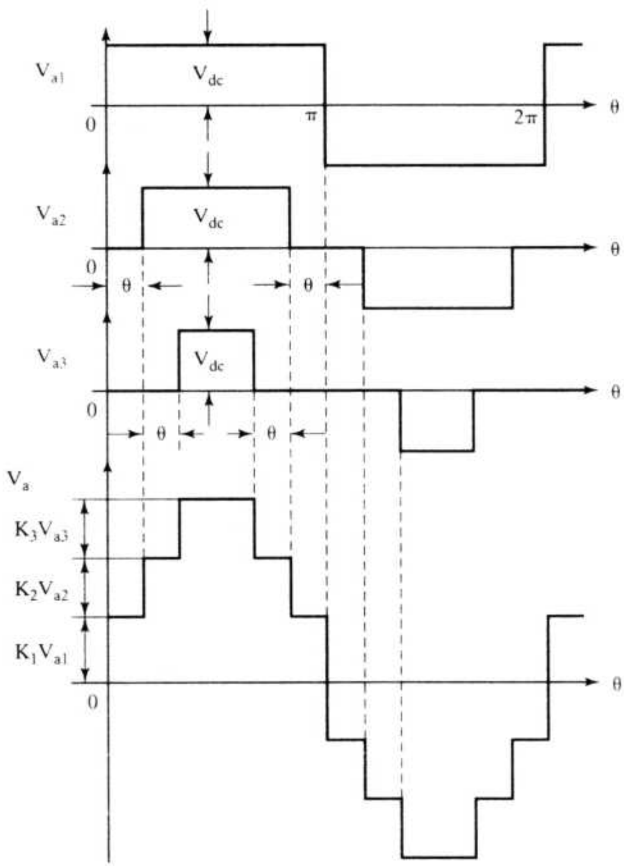
7.4.9.1 General. Undesirable pulsating torques and additional losses in the induction motor are generated by harmonic input voltages. Some of the dominant harmonics can be eliminated selectively by waveshaping the inverter output voltages. It can be done either by phase-shifting two or more inverters and summing their outputs or by pulse-width-modulating the output voltages. Both approaches are described in this section.

7.4.9.2 Phase-shifting control. If the output voltages of two or more inverters fed from a common dc source are phase-shifted and summed through a set of output transformers, a multistep voltage can be generated. The amount of phase shift and magnitude of individual inverter voltages determine the harmonic contents of the resultant voltages. For example, consider the inverters and their waveforms shown in Figure 7.35. The three inverter voltages are equally spaced from each other and summed with different gains through a transformer. The turns ratios of the transformers are chosen to be

$$K_1 = 1$$
$$K_2 = 2\cos\theta \tag{7.157}$$



(i) Inverter arrangement



(ii) Summation of voltages

Figure 7.35 Stepped-waveform generation

$$K_3 = 2\cos 2\theta$$

It is generalized for x inverters as

$$K_x = 2\cos(x-1)\theta, \quad x \geq 2 \tag{7.158}$$

where θ is the phase shift of the inverter voltages, defined as

$$\theta = \frac{2\pi}{(\text{steps in the waveforms})} = \frac{2\pi}{n} = \frac{2\pi}{4x} = \frac{\pi}{2x} \tag{7.159}$$

where n is the number of steps in the voltage. Each inverter introduces 4 steps; therefore, x inverters introduce $4x$ steps into the output voltage. The output of the fundamental voltage for three steps is obtained as

$$V_{a1} = 4\frac{V_{dc}}{\pi}[K_1 + K_2\cos\theta + K_3\cos 2\theta]\sin\omega_s t \tag{7.160}$$

Substituting for the turns ratio from equation (7.157) into (7.160) gives the fundamental voltage as

$$V_{a1} = 12\frac{V_{dc}}{\pi}\sin\omega_s t \tag{7.161}$$

Similarly, the fifth and seventh harmonic voltages are

$$V_{a5} = \frac{4V_{dc}}{5\pi}[K_1 + K_2\cos 5\theta + K_3\cos 10\theta]\sin 5\omega_s t = 0 \tag{7.162}$$

$$V_{a7} = \frac{4V_{dc}}{7\pi}[K_1 + K_2\cos 7\theta + K_3\cos 14\theta]\sin 7\omega_s t = 0 \tag{7.163}$$

The eleventh and thirteenth harmonic voltages are,

$$V_{a11} = \frac{V_{a1}}{11} \tag{7.164}$$

$$V_{a13} = \frac{V_{a1}}{13} \tag{7.165}$$

Note that there is a complete cancellation of harmonics in the case of the fifth and seventh. Hence, phase-shifting neutralizes harmonics below the sideband frequencies of the n-step voltages. In this case, the minimum sideband frequencies are $n - 1$ and $n + 1$, i.e., 11 and 13.

The voltage magnitude of the multistepped waveforms is varied by controlling the magnitude of dc source voltage. This has the advantage of keeping the harmonic contents to a minimum. Also, the voltage magnitude can be controlled by phase-shifting the various inverter voltages instead of keeping it a constant, but the total harmonic distortion will increase with phase-shifting.

This technique of phase-shifting and neutralizing the undesirable harmonics involves the use of transformers at the output stage that increases the cost and space requirements of the inverter drive, but it has the advantage of paralleling many inverters of small capacities to obtain a larger inverter capability. This technique avoids also the paralleling of devices to obtain higher capacity.

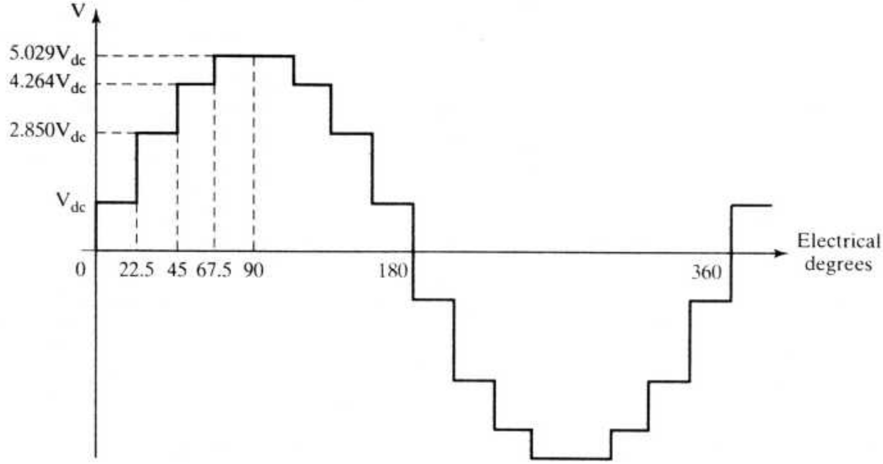


Figure 7.36 Sixteen-stepped waveform to neutralize harmonics up to the fifteenth

Example 7.7

Using the phase-shifting principle, find the number of inverters, their phase shifts, and the respective turns ratio to suppress harmonics lower than the fifteenth.

Solution To suppress harmonics lower than the fifteenth amounts to an acceptable minimum sideband frequency of $n - 1$, where n is the number of steps in the voltage waveform.

(i) Therefore, $n = 15 + 1 = 16$ steps

(ii) Number of inverters: $x = n/4 = 16/4 = 4$

(iii) Phase shift: $\theta = \frac{360°}{n} = \frac{360°}{16} = 22.5°$

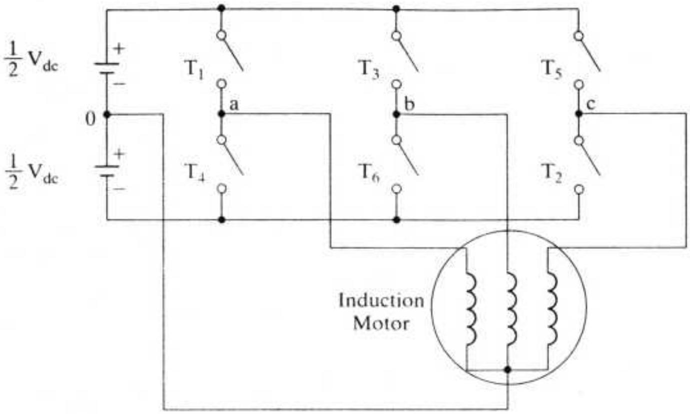
(iv) $K_1 = 1$

$$K_2 = 2\cos\theta = 2\cos 22.5° = 1.847$$
$$K_3 = 2\cos 2\theta = 2\cos 45° = 1.414$$
$$K_4 = 2\cos 3\theta = 2\cos 67.5° = 0.765$$

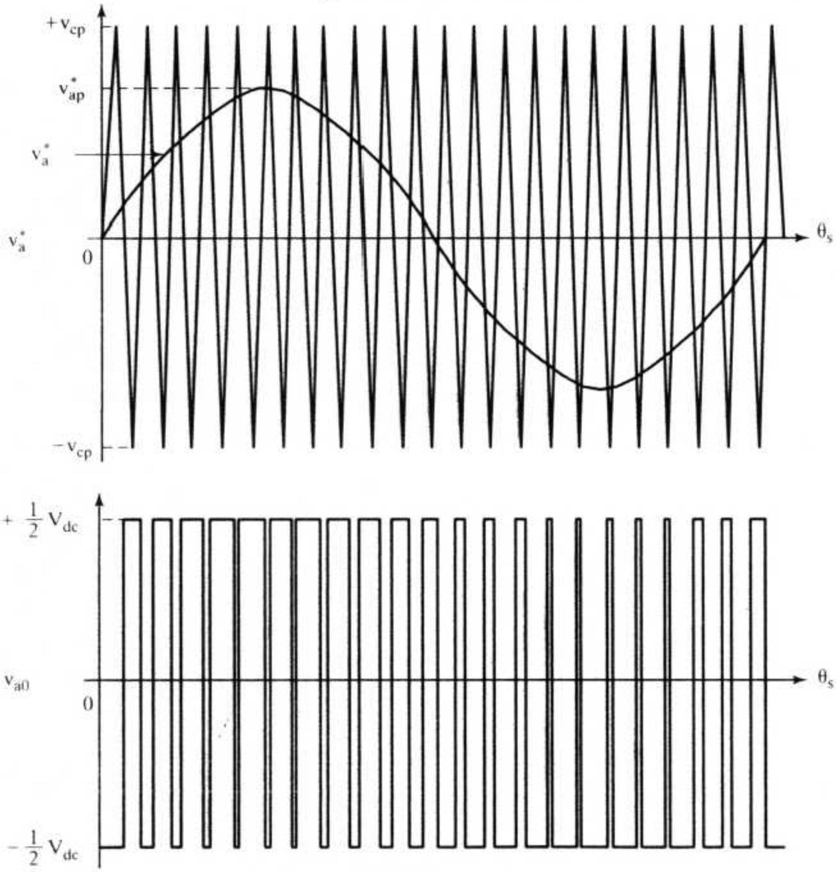
The output-voltage waveform is shown in Figure 7.36.

7.4.9.3 Pulse-width modulation (PWM). The control of harmonics and variation of the fundamental component can be achieved by chopping the input voltage. A number of pulse-width-modulation schemes have been in use in the inverter-fed induction motor drives. All of these PWM schemes aim to maximize the fundamental and selectively eliminate a few lower harmonics. Some are discussed in this section.

Figure 7.37 shows the schematic of an inverter and waveforms for a phase mid-pole voltage. The intersections of the carrier signal v_c (usually a bidirectional triangular



(i) Inverter motor schematic



(ii) PWM operation

Figure 7.37 Sinusoidal pulse-width modulation waveform) and the commanded fundamental v_a^* (usually a sinusoidal waveform) provide the switching signals to the base drive of the inverter switches. The switching logic for one phase is summarized as

$$\begin{aligned} v_{a0} &= \frac{1}{2} V_{dc}, \quad v_c < v_a^* \\ &= -\frac{1}{2} V_{dc}, \quad v_c > v_a^* \end{aligned} \tag{7.166}$$

The fundamental of this midpoint voltage is

$$v_{a01} = \frac{V_{dc}}{2} \cdot \frac{v_{ap}^*}{v_{cp}} \tag{7.167}$$

where v_{ap}^* is the peak value of the *a* phase reference or command signal and v_c is the peak value of the triangular carrier signal.

The resulting output has the frequency and phase of the reference signal. The modulation index or ratio is defined by

$$m = \frac{v_{ap}^*}{v_{cp}} \tag{7.168}$$

which, substituted into equation (7.167), gives

$$v_{a01} = m \frac{V_{dc}}{2} \tag{7.169}$$

Varying modulation index changes the fundamental amplitude, and varying the frequency of reference v_a^* changes the output frequency. The ratio between the carrier and reference frequencies, f_c/f_s, changes the harmonics. To eliminate a large number of lower harmonics, f_c has to be very high. Note that this will entail high switching losses and a considerable derating of the supply voltage, because having many turn-on and turn-off intervals sizably reduces the voltage available for output. It is usual to have a fixed value for f_c as say from 9 up to the base frequency of the induction motor, and then reduced values of f_c as higher-harmonic torque pulsations do not significantly affect the drive performance at high speed. The carrier and reference signals have to be synchronized to eliminate the beat-frequency voltage appearing at the output. If the f_c ratio is very high, then synchronization is not very critical. A typical relationship between the carrier and reference frequencies is shown in Figure 7.38. At low frequencies, less than 40 Hz, the carrier frequency could be fixed or a variable to have the synchronization feature.

An alternative sinusoidal pulse-width-modulation strategy is shown in Figure 7.39. The reference contains the fundamental and third harmonic. As the third harmonic is canceled in a star-connected system, the fundamental is enriched in this strategy.

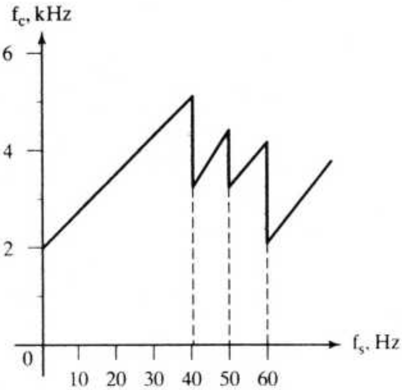


Figure 7.38 Relationship between carrier and fundamental stator frequency

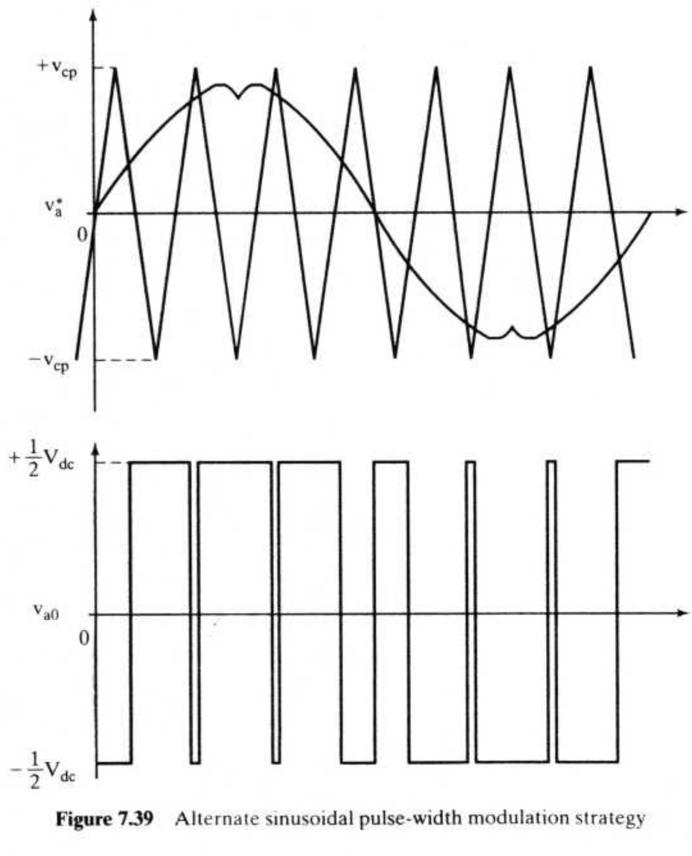


Figure 7.39 Alternate sinusoidal pulse-width modulation strategy

7.4.10 Steady-State Evaluation with PWM Voltages

Irrespective of the control strategies employed in the induction motor drive, the input voltages are periodic in steady state. Hence, direct steady-state performance evaluation is possible by matching boundary conditions. In this section, PWM voltage inputs are considered for steady-state performance evaluation of the induction motor drive system. The PWM can be generated in any number of ways: sine-triangle, trapezoidal-triangle, space vector, sampled asymmetric method modulation strategies, etc. Because of their symmetry for either half-wave or full-wave, the boundary-matching technique is ideal for evaluating the steady-state current vector directly, without going through the dynamic simulation from start-up. The algorithm for this follows and is illustrated with three-phase sampled asymmetric modulated voltage inputs to the induction motor whose parameters are given in Example 7.1.

7.4.10.1 PWM voltage generation. The PWM voltages are generated from the sequences of pulses whose turn-on times are given by the expression

$$t(i) = \frac{1 \pm m \sin[a(i)]}{2f_c}, \qquad \begin{array}{l} + \text{ for even value of n} \\ - \text{ for odd value of n} \end{array} \tag{7.170}$$

where n is the ratio between the carrier and modulation frequencies, t(i) is the ith pulse width, m is the modulation ratio, f_c is the carrier frequency, and

$$a(i) = \frac{2\pi i}{n}, \text{ rad}; \qquad i = 1, 2, ..., n \tag{7.171}$$

The pulse widths are spread equally on either side of the pulse centers given by the expression,

$$p_c(i) = \frac{2i - 1}{2}\frac{2\pi}{n} \tag{7.172}$$

The pulse widths t(i) could be expressed in electrical radians as

$$p_w(i) = t(i)f_s, \text{ rad} \tag{7.173}$$

where f_s is the modulation frequency, i.e., the fundamental frequency desired for motor input voltages. The positions of the pulses are found from $p_c(i)$ and $p_w(i)$. During on-time, the midpole voltage is half of the dc link voltage, $0.5V_{dc}$; during off-time, the midpole voltage is negative half of the dc link voltage, $-0.5V_{dc}$. From the midpole voltages, the line voltages and, in turn from them, the phase voltages are derived by using equations (7.8) to (7.10). In a voltage-source inverter, note that complementary switching in each phase leg is used to provide for a definite predetermined voltage across the load irrespective of the load current.

Further, in digital implementations, the pulse widths are approximated, depending on the number of bits involved in the digital controller. For illustration, the following parameters are chosen:

$$m = 0.5, f_s = 60 \text{ Hz}, n = 9, f_c = nf_s = 540 \text{ Hz}, V_b = 163.3 \text{ V}$$

The pulse widths and their approximations for implementation are given in the following table

i	$p_w(i)$, deg.	approximated $p_w(i)$, deg.
1	20	20
2	26.43	26
3	29.85	30
4	28.66	29
5	23.41	23
6	16.57	17
7	11.33	11
8	10.16	10
9	13.59	14

For the direct steady-state evaluation of the current vector and hence of the performance of the induction motor drive, the PWM phase and *d* and *q* axes voltages in stator reference frames are obtained in p.u., as is shown in Figure 7.40. Note that the *d* and *q* axes voltages are obtained from the following relationships:

$$v_{qs} = \frac{2}{3}[v_{as} - 0.5(v_{bs} + v_{cs})] \tag{7.174}$$

$$v_{ds} = \frac{1}{\sqrt{3}}[v_{cs} - v_{bs}] \tag{7.175}$$

and the zero-sequence component is zero because the set of voltages is balanced. The next step, then, is to use these voltages in the machine model to obtain the steady-state current vector.

7.4.10.2 Machine model. The induction-machine model in the stator reference frames can be written in the following form:

$$V = (R + Lp)i + G\omega_r i \tag{7.176}$$

where

$$V = [v_{as} \quad v_{ds} \quad 0 \quad 0]^t \tag{7.177}$$

$$i = [i_{qs} \quad i_{ds} \quad i_{qr} \quad i_{dr}]^t \tag{7.178}$$

$$R = \begin{bmatrix} R_s & 0 & 0 & 0 \\ 0 & R_s & 0 & 0 \\ 0 & 0 & R_r & 0 \\ 0 & 0 & 0 & R_r \end{bmatrix} \tag{7.179}$$

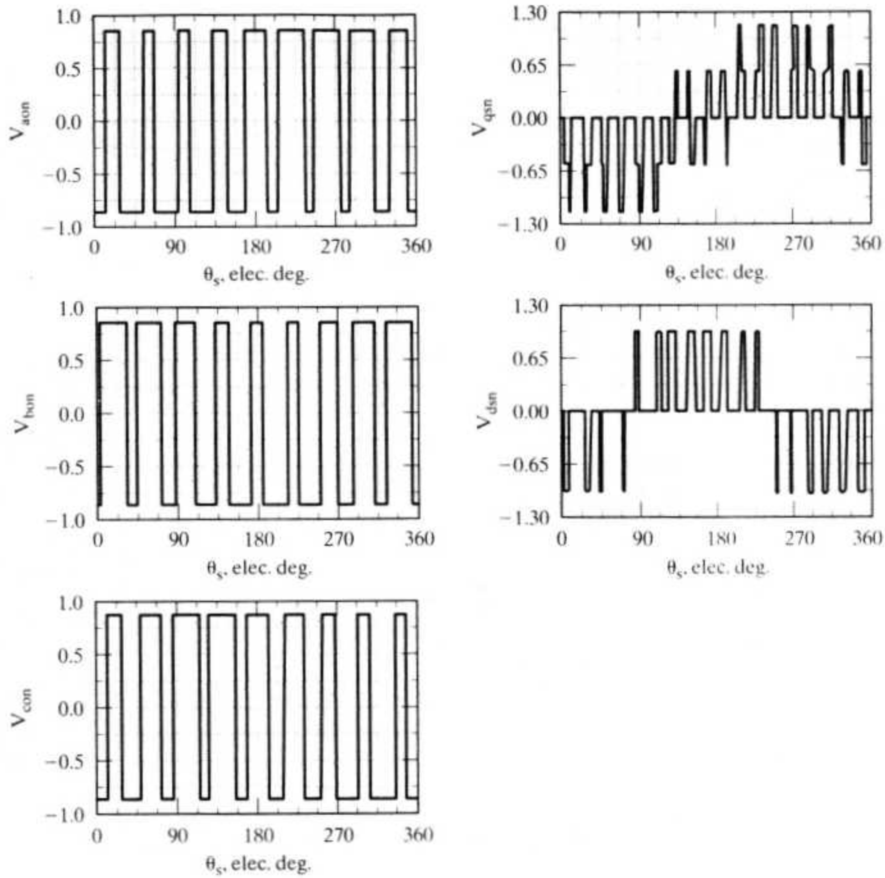


Figure 7.40 PWM voltages in *abc* and *dq* frames in p.u.

$$L = \begin{bmatrix} L_s & 0 & L_m & 0 \\ 0 & L_s & 0 & L_m \\ L_m & 0 & L_r & 0 \\ 0 & L_m & 0 & L_r \end{bmatrix} \tag{7.180}$$

$$G = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -L_m & 0 & -L_r \\ L_m & 0 & L_r & 0 \end{bmatrix} \tag{7.181}$$

which is cast in state-space form as

$$\dot{X} = AX + Bu \tag{7.182}$$

where

$$A = -L^{-1}[R + \omega_r G] \tag{7.183}$$

$$B = L^{-1} \tag{7.184}$$

$$X = i \tag{7.185}$$

$$u = V \tag{7.186}$$

The next step is to use this set of state-space equations to obtain the steady-state current vector directly.

7.4.10.3 Direct evaluation of steady-state current vector by boundary-matching technique. The state-space equations can be discretized as follows. The solution for the current vector is

$$X(t) = e^{At}X(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau \tag{7.187}$$

In one sampling interval, T_s, the input and state variables are constant, by discretization. Hence,

$$X(T_s) = e^{AT_s}X(0) + e^{AT_s}\int_0^{T_s} e^{-A\tau}d\tau Bu(0) \tag{7.188}$$

which gives rise to a solution of the form

$$X(T_s) = e^{AT_s}X(0) + (e^{AT_s} - I)A^{-1}Bu(0) \tag{7.189}$$

and it could be generalized for the k^{th} sampling interval (and omitting the T_s in the parentheses for simplicity):

$$X(k + 1) = \Phi X(k) + Fu(k) \tag{7.190}$$

where

$$\Phi = e^{AT_s} \text{ and } F = (\Phi - I)A^{-1}B \tag{7.191}$$

Note that $X(k)$ can be calculated in terms of $X(k-1)$ and $u(k-1)$ and so on, as in the following:

$$X(1) = \Phi X(0) + Fu(0) \tag{7.192}$$

$$X(2) = \Phi X(1) + Fu(1) \tag{7.193}$$

and, substituting for $X(1)$ from the previous relationship and expanding the equation,

$$X(2) = \Phi^2X(0) + \Phi Fu(0) + Fu(1) \tag{7.194}$$

and similarly, for the $(k+1)$th sampling interval, the current vector is

$$X(k + 1) = \Phi^{k+1}X(0) + \Phi^kFu(0) + \Phi^{k-1}Fu(1) + \ldots + Fu(k) \tag{7.195}$$

The last expression, by symmetry of the wave forms, must be equal to the initial vector itself if the $(k+1)$th sampling interval corresponds to 360 electrical degrees. Equating these, we get

$$X(k + 1) = X(0) \tag{7.196}$$

but $X(k+1)$ contains the term $X(0)$; rearranging these, the steady-state initial vector $X(0)$ is obtained as

$$\begin{aligned} X(0) &= [I - \Phi^{(k+1)}]^{-1}\{\Phi^k Fu(0) + \Phi^{k-1}Fu(1) + \ldots + Fu(k)\} \\ &= [I - \Phi^{(k+1)}]^{-1}\left\{\sum_{j=0..k} \Phi^j Fu(k - j)\right\} \end{aligned} \tag{7.197}$$

where I is the 4×4 identity matrix. For the example under consideration, the sampling time corresponds to 2 electrical degrees, given as

$$T_s = \frac{2}{360f_s}, s \tag{7.198}$$

The exponential of AT_s can be evaluated from the series with 8 to 15 terms. The value of k is given by

$$k + 1 = 360/2 = 180 \tag{7.199}$$

which gives $k = 179$. For wave forms with half-wave symmetry, the final and initial values are related by other than identity, as is shown in the section on the six-step inverter-fed induction motor drive.

7.4.10.4 Computation of steady-state performance. The current vector is evaluated for the entire cycle from the discretized state-space equation discussed and derived above. The electromagnetic torque is computed as

$$T_e(k) = \frac{3}{2}\frac{P}{2} L_m \{i_{qs}(k)i_{dr}(k) - i_{ds}(k)i_{qr}(k)\} \tag{7.200}$$

where the currents are the components of the state vector $X(k)$. By inverse transformation, the phase currents are computed as

$$i_{as}(k) = i_{qs}(k) \tag{7.201}$$

$$i_{bs}(k) = -0.5i_{qs}(k) - 0.866i_{ds}(k) \tag{7.202}$$

The stator q and d axes currents, a and b phase currents, and electromagnetic torque are shown in Figure 7.41 for the induction machine whose parameters are given in Example 7.1, but running with a slip of 0.05 for the present illustration. A significant difference between these current wave forms and those of the six-step inverter-fed induction motor drives is to be noted: the currents have become more sinusoidal, with the superposed switching ripples. Further, the rate of change of currents and sharpness of the peaks have become smaller by comparison, contributing to lower overall rms current, resulting in lower stator copper losses and better thermal performance of the machine. Even though the switching harmonic torque has higher magnitude, note that its frequency has increased in proportion to the PWM frequency, indicating that it is easier to filter them with the mechanical inertia of the machine and its load than that of the sixth-harmonic ripple present in the six-step inverter-fed machines.

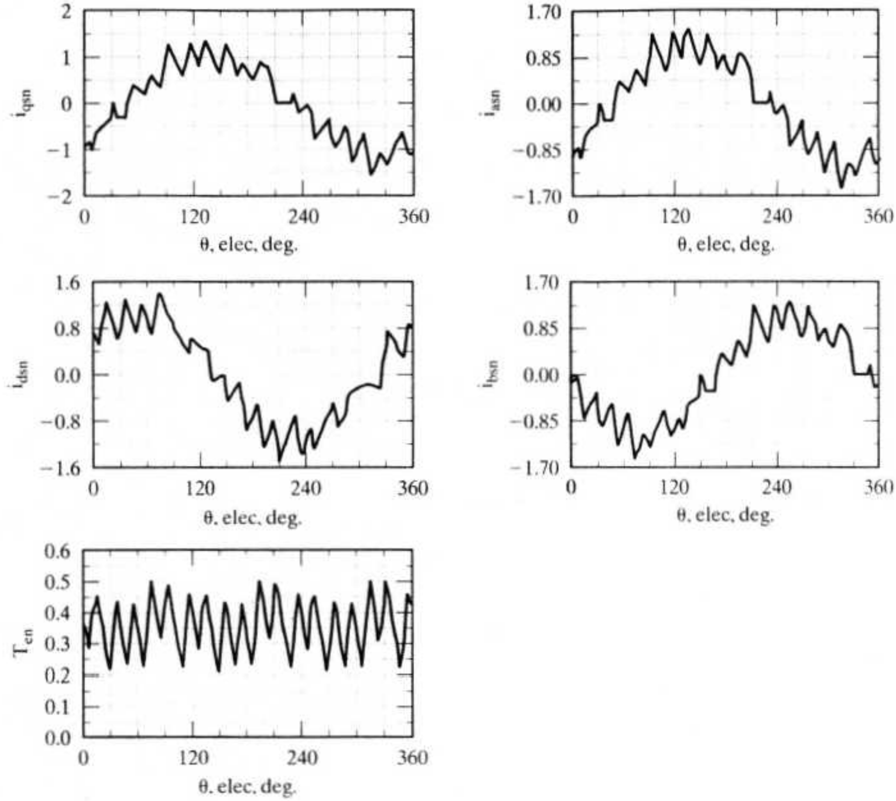


Figure 7.41 Direct steady-state evaluation of PWM-fed induction motor drive in p.u.

Example 7.8

Compute the transient response of the PWM-based induction motor drive with volts/Hz control strategy. The motor and PWM details are as follows:

100 hp, 460 V, 6 pole, 60 Hz, 3 phase, star connected induction motor.

Efficiency = 0.82, Power factor = 0.86, $R_s = 0.0551\ \Omega$, $R_r = 0.311\ \Omega$, $L_m = 0.02066$ H, $L_{ls} = 0.0007798$ H, $L_{lr} = 0.00007798$ H, $J = 0.8$ kg-m^2, $B_1 = 0.15$ N·m/(rad/sec).

$$\text{PWM: } \frac{f_c}{f_s} = 9.$$

Solution The steady state of the PWM-fed induction motor was calculated with the preset PWM voltages given in equations (7.170) to (7.173). Alternately, the PWM voltages can be generated from the intersections of the carrier and modulation signals. The transient response of such a PWM-based induction motor drive with volts/Hz control strategy and on open-loop speed control is computed by using the dynamic model of the induction motor. The offset voltage, V_o, and the volts-to-frequency constant, K_{vf}, are calculated by using stator full-

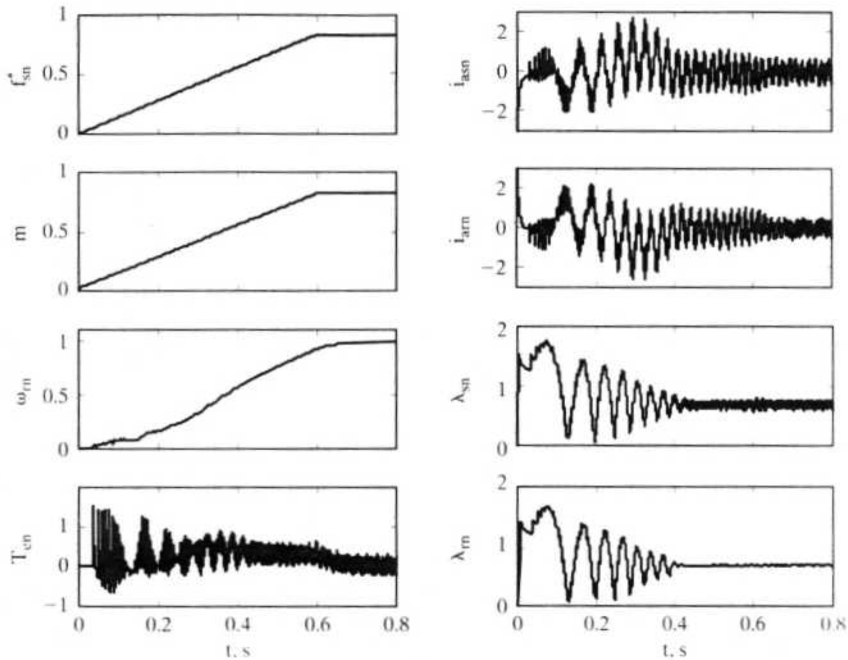


Figure 7.42(i) Starting transient response of the PWM-based open-loop volts/Hz induction motor drive in normalized units.

load current. A ramp frequency command is given, which serves as a soft start, and the modulation ratio is calculated as

$$m = \frac{2}{V_{dc}}(V_o + K_{vf}f_s)$$

The phase command voltage is also given as

$$v_{as}^* = mv_c \sin(\omega_s t)$$

where v_c is the absolute peak carrier voltage and in this case is taken to be 10 V. The intersection of the carrier and the command voltages is obtained by sampling every microsecond. Then the midpole voltages, i.e., the midpoints of dc link voltage and the midpoint of inverter phases are obtained from the intersections. The midpoint voltages are $\pm \frac{V_{dc}}{2}$, where V_{dc} is the dc link voltage.

In this example, it is assumed that $V_{dc} = \sqrt{2} \times 460$ V.

$$V_b = \sqrt{2} \times \frac{460}{\sqrt{3}}\text{V}, T_b = \frac{P}{2}\frac{P_b}{\omega_b}, P_b = 100 \text{ hp}, f_b = 60 \text{ Hz}, I_b = \frac{2}{3}\frac{P_b}{V_b}, \omega_b = 2\pi f_b.$$

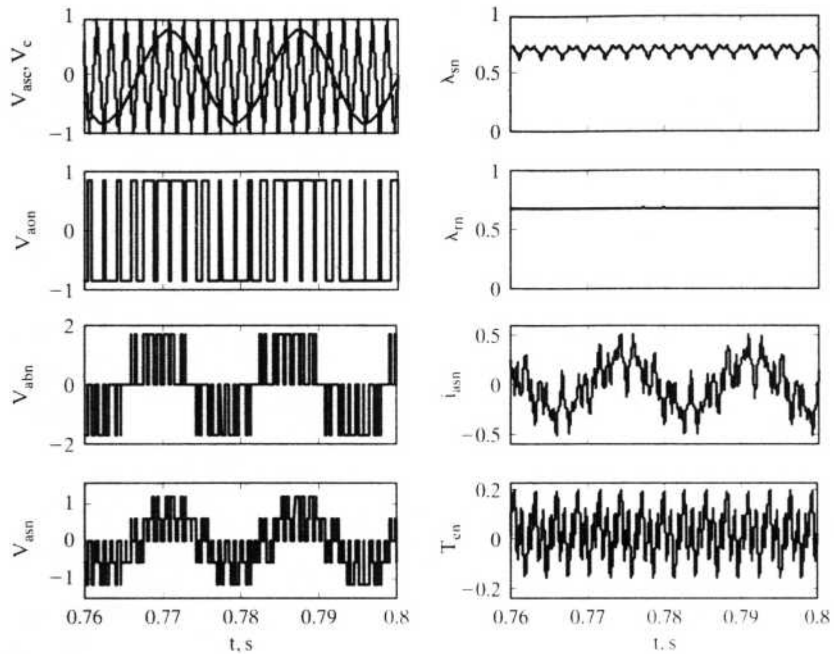


Figure 7.42(ii) Steady-state response of the PWM-based open-loop volts/Hz induction motor drive

From the midpole voltages, the line voltages are calculated as

$$V_{ab} = V_{ao} - V_{bo}$$
$$V_{bc} = V_{bo} - V_{co}$$
$$V_{ca} = V_{co} - V_{ao}$$

from which the phase voltages are derived as

$$V_{as} = \frac{V_{ab} - V_{ca}}{3}$$

$$V_{bs} = \frac{V_{bc} - V_{ab}}{3}$$

$$V_{cs} = \frac{V_{ca} - V_{bc}}{3}$$

The *dq* voltages are derived from the phase voltages in synchronous reference frames and the motor equations are integrated to obtain the currents and torque. The *abc* currents are obtained from the synchronous-reference-frame *dq* currents by the inverse transformation. The transient response for the start-up is shown in Figure 7.42(i). The modulation ratio follows the command frequency with the adjusted magnitude. The stator and rotor current magnitudes are smaller than those of the six-step inverter-fed volts/Hz-controlled drive. The soft start helps this reduc-

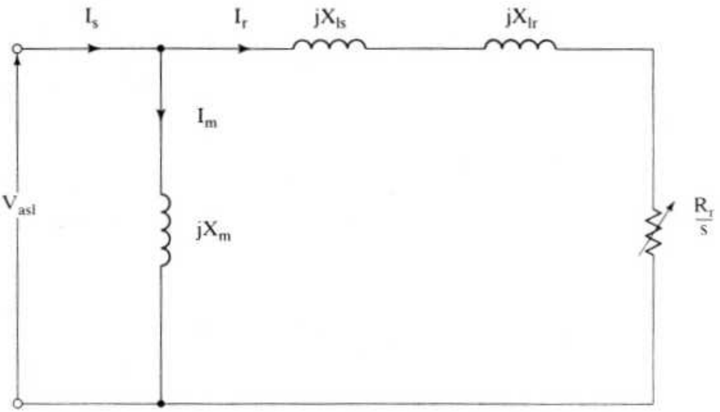


Figure 7.43 Simplified equivalent circuit of the induction motor

tion in current magnitude. The stator and rotor flux linkages reach steady state at around 0.5 s. The rotor torque has large oscillations, caused mainly by the stator and rotor flux-linkage oscillations. The rotor speed has a slow rise until the flux linkages are settled, and then the speed rises smoothly.

When the motor has reached steady state, the waveforms for a few cycles are shown in Figure 7.42(ii). These waveforms show the generation of PWM signals, the midpole voltages, a line-to-line and a phase voltage, a stator phase current, stator and rotor flux linkages, and electromagnetic torque. The motor drive is operating at no-load but overcomes the friction and hence requires only a small average torque as shown in the figure. All the variables in Figures 7.42(i) and (ii) are in normalized units.

7.4.11 Flux-Weakening Operation

7.4.11.1 Flux weakening. The air gap flux of the induction motor is maintained constant by keeping the ratio between the induced air gap emf and stator frequency a constant. The voltage input to the inverter is limited, and, at its maximum value, it is designed to give rated flux operation at rated frequency of the induction motor. For operation at higher than rated frequency, the dc voltage input to the inverter is clamped and thereafter the induction motor is fed with a constant voltage. This leads to the flux weakening. Consider the induction-motor equivalent circuit given in Figure 7.43. Stator resistance is neglected in this equivalent circuit. The air gap flux linkage is

$$\lambda_m = \frac{V_{as1}}{\omega_s} \tag{7.203}$$

With increasing stator frequency, the air gap flux linkage is decreasing. This has an interesting consequence on the performance of the motor drive. Very similar to the dc motor drives, the ac motors are also flux-weakened for operation above rated speed. At high speed, the output of the motor will be exceeded if the torque is not programmed to vary inversely proportionally to the speed. The torque is programmed as

$$T_e \propto \frac{1}{\omega_m} \tag{7.204}$$

or

$$T_e = \frac{\text{rated output}}{\omega_m} = \frac{P_m}{\omega_m} \tag{7.205}$$

where P_m is the output power.

Equation (7.205) is further written as

$$T_e = \frac{P_m}{\omega_s(1 - s)} \cdot \frac{P}{2} \tag{7.206}$$

By writing electromagnetic torque in terms of air gap flux linkage, voltage, and motor parameters, and by using a simple equivalent circuit with X_m being very large, we get

$$T_e = 3\frac{P}{2} \cdot \frac{V_{as1}^2}{\left(\frac{R_r}{s}\right)^2 + X_{eq}^2} \cdot \left(\frac{R_r}{s\omega_s}\right) = 3\frac{P}{2} \cdot \lambda_m \cdot \omega_s \cdot \frac{V_{as1}}{\left(\frac{R_r}{s}\right)^2 + X_{eq}^2} \cdot \left(\frac{R_r}{s\omega_s}\right)$$

$$= 3\frac{P}{2} \cdot \lambda_m \cdot \frac{V_{as1}}{\left(\frac{R_r}{s}\right)^2 + X_{eq}^2} \cdot \left(\frac{R_r}{s}\right) \tag{7.207}$$

where

$$X_{eq} = X_{lr} + X_{ls} \tag{7.208}$$

Equating (7.206) and (7.207), we get

$$P_m = 3\lambda_m \cdot \omega_s \cdot \frac{V_{as1}}{\left(\frac{R_r}{s}\right)^2 + X_{eq}^2} \cdot \left(\frac{R_r}{s}\right)(1 - s) \tag{7.209}$$

From equation (7.209), it is inferred that, to maintain the air gap power constant, the air gap flux linkage has to be varied inversely proportionally to the stator frequency. The stator voltage input is a constant during flux weakening, so, if slip s is small, then the air gap power becomes a constant, but maintaining air gap power constant does not guarantee that the shaft power output is constant. To do so, the slip also has to be regulated. The flux-weakening and hence torque-programming is inherent in the induction motor by simple variation of stator frequency, and this makes the induction motor drive attractive in high-speed applications. This is true only in an approximate sense. An accurate control and maintenance of constant shaft power output during flux-weakening is an involved process, taken up in detail in Chapter 8.

7.4.11.2 Calculation of slip. Because the input phase voltage is a constant, the induction motor is controlled by varying its slip speed. For a given air gap power, the slip is obtained as follows.

The electromagnetic torque is given by

$$T_e = 3\frac{V_{as1}^2}{\left(\frac{R_r}{s}\right)^2 + L_{eq}^2\omega_s^2}\cdot\left(\frac{R_r}{s\omega_s}\right) \tag{7.210}$$

The rated value or torque is

$$T_{er} = 3\frac{V_{as1}^2}{\left(\frac{R_r}{s_r}\right)^2 + L_{eq}^2\omega_{sr}^2}\cdot\left(\frac{R_r}{s\omega_{sr}}\right) \tag{7.211}$$

where the subscript *r* denotes the rated value of the corresponding variables.

The normalized torque is

$$T_{en} = \frac{T_e}{T_{er}} = \left(\frac{s}{s_r}\right)\left(\frac{1}{\omega_{sn}}\right)\frac{(R_r^2 + s_r^2X_{eq}^2)}{(R_r^2 + s^2\omega_{sn}^2X_{eq}^2)} \tag{7.212}$$

where

$$\omega_{sn} = \frac{\omega_s}{\omega_{sr}} \tag{7.213}$$

and ω_{sn} is the normalized stator frequency. (Its maximum value is usually a fixed value at the design stage of the inverter.)

Also, the normalized torque in the field-weakening mode is given by

$$T_{en} = \frac{P_{mn}}{\omega_{mn}} = \frac{P_{mn}}{(\omega_m/\omega_{mr})} = P_{mn}\frac{(1-s_r)}{(1-s)}\cdot\frac{\omega_{sr}}{\omega_s} = P_{mn}\frac{(1-s_r)}{(1-s)}\cdot\frac{1}{\omega_{sn}}, \text{ p.u.} \tag{7.214}$$

where P_{mn} is the normalized output power in p.u.

Equating (7.212) and (7.214), we get

$$P_{mn}\frac{(1-s_r)s_r}{R_r^2 + s_r^2X_{eq}^2} = \frac{(1-s)s}{R_r^2 + s^2\omega_{sn}^2X_{eq}^2} \tag{7.215}$$

By noting that the left-hand side of equation (7.214) is a constant and by defining

$$a = \frac{R_r}{X_{eqr}} \tag{7.216}$$

$$b = \frac{P_{mn}(1-s_r)s_r}{a^2 + s_r^2} \tag{7.217}$$

equation (7.217) is written as

$$\frac{(1-s)s}{a^2 + s^2\omega_{sn}^2} = b \tag{7.218}$$

from which the slip is solved for as

$$s = \frac{1}{2(1 + b\omega_{sn}^2)} \pm \frac{1}{2}\sqrt{\frac{1}{(1 + b\omega_{sn}^2)^2} - \frac{4ba^2}{(1 + b\omega_{sn}^2)}} \qquad (7.219)$$

Note that in open-loop operation, this slip must be less than the slip corresponding to the breakdown torque or peak torque, denoted as s_m. The constraint for open-loop operation, then, is

$$s < s_m \qquad (7.220)$$

Usually, this is accompanied by another constraint, that is, the stator current should not exceed a certain value. In most cases, this is the rated value of the stator current, denoted by I_{sr}. Then the constraint is expressed as

$$I_s < I_{sr} \qquad (7.221)$$

where

$$I_s = \frac{V_{as1}}{\left(\frac{R_r}{s}\right) + jL_{eq}\omega_s} + \frac{V_{as1}}{j\omega_s L_m} \qquad (7.222)$$

Equation (7.219) for slip gives two values, the smaller value being the statically stable region of the torque–speed characteristics and the larger value corresponding to the operating point in the unstable region of the torque–speed characteristics. For regeneration, the rated slip s_r has to be negative, and hence the slip calculated from equation (7.219) will yield negative values; P_{mn} will be negative for this condition.

7.4.11.3 Maximum stator frequency. The maximum operating stator frequency for the motor drive is evaluated from the fact that the slip cannot be a complex value; hence, from equation (7.219),

$$\frac{1}{(1 + b\omega_{sn}^2)^2} \geq \frac{4a^2b}{(1 + b\omega_{sn}^2)} \qquad (7.223)$$

The maximum stator frequency is

$$\omega_{sn(max)} = \frac{\sqrt{1 - 4a^2b}}{2ab}, \text{ p.u.} \qquad (7.224)$$

from which the maximum rotor speed is obtained as

$$\omega_{mn(max)} = \omega_{sn(max)} \cdot \frac{(1 - s)}{(1 - s_r)}, \text{ p.u.} \qquad (7.225)$$

Example 7.9

Calculate the stator current magnitude and slip at maximum stator frequency when the induction motor drive is in the flux-weakening mode of operation and delivering rated power. The motor and drive details are given in Example 7.1.

Solution Air gap power is assumed to be equal to rated value.

$$s_r = 0.0183$$

$$a = \frac{R_r}{X_{eq}} = \frac{0.183}{(0.554 + 0.841)} = 0.131$$

$$b = P_{mn}\frac{(1 - s_r)s_r}{a^2 + s_r^2} = \frac{1(1 - 0.0183)(0.0183)}{(0.131)^2 + (0.0183)^2} = 1.024$$

$$\omega_{sn(max)} = \frac{\sqrt{1 - 4a^2b}}{2ab} = \frac{\sqrt{1 - 4 \times 0.131^2 \times 1.024}}{2 \times 0.131 \times 1.024} = 3.59 \text{ p.u.}$$

$$s = \frac{1}{2(1 + b\omega_{sn}^2)} + \frac{1}{2}\sqrt{\frac{1}{(1 + b\omega_{sn}^2)^2} - \frac{4a^2b}{(1 + b\omega_{sn}^2)}} = \frac{1}{2(1 + 1.024 \times 3.59^2)} = 0.0359 \text{ p.u.}$$

$$\omega_s = \omega_{sn} \times \omega_{sr} = 3.59 \times 377 = 1353.4 \text{ rad/sec}$$

$$V_{as1} = 115.5 \text{ V}$$

$$I_s = V_{as1}\left\{\frac{1}{\left(\frac{R_r}{s}\right) + jL_{eq}\omega_s} + \frac{1}{j\omega_s L_m}\right\} = 115.5\left\{\frac{1}{\frac{0.183}{0.0183} + j5.0} + \frac{1}{j1353.4 \times 0.0538}\right\}$$

$$= 9.24 - j6.2 \text{ A}$$

$$|I_s| = 11.12 \text{ A}$$

$$\omega_{mn(max)} = \omega_{sn(max)} \cdot \frac{(1 - s)}{(1 - s_r)} = 3.59 \times \frac{(1 - 0.0359)}{(1 - 0.0183)} = 3.525 \text{ p.u.}$$

7.5 CURRENT-SOURCE INDUCTION MOTOR DRIVES

7.5.1 General

In a current-source drive, the input currents are six-stepped waveforms. Amplitude and frequency are variables, as in voltage-source drives. Current-source drives have a distinct advantage over voltage-source drives: the electrical apparatus is current-sensitive, and torque is directly related to the current rather than the voltage. Hence, control of current ensures the direct and precise control of the electromagnetic torque and drive dynamics. Current-source variable-frequency supplies are realized either with an autosequentially commutated inverter (hereafter referred to as ASCI) or with current-regulated inverter drives. Their operation is explained in the following sections. The closed-loop four-quadrant current-source inverter-fed induction motor and its performance are discussed. The method of computing steady-state and dynamic per-

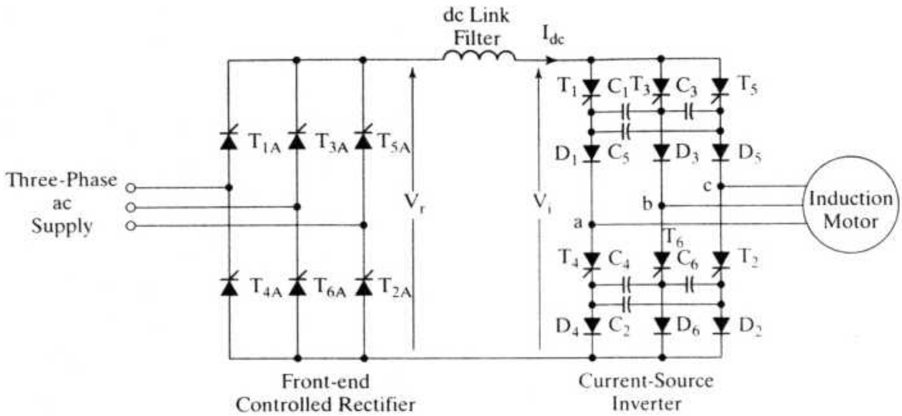


Figure 7.44 Current-source induction motor drive

formance is given. The effect of lower harmonics on the performance of the motor drive is evaluated, and control of harmonics is explained in this section.

7.5.2 ASCI

The ASCI-fed induction motor drive is shown in Figure 7.44. The converter system has a controlled rectifier for providing the ac-to-dc conversion and an inverter for dc-to-ac power conversion. The dc output voltage is fed to the autosequentially commutated current-source inverter (ASCI) through a filter inductor. This inductor is provided to maintain the dc link current at a steady value. The operation of the controlled rectifier and current source is given in Chapter 3. The principle of operation, to facilitate the understanding of the current-source drive, is given below.

7.5.2.1 Commutation. The sequence of firing the ASCI is T_1, T_2, T_3, T_4, T_5, T_6 for the phase sequence *abc* in the induction motor. At any time, two SCRs are conducting, and they are turned on at an interval of 60 electrical degrees. Let T_1 and T_2 conduct, and let the dc link current be a constant. The capacitor C_1 is charged positive at the plate connecting it to the cathode of T_1 and negative at the plate connecting it to the cathode of T_3. The current is following the path T_1, D_1, phase *a* winding, phase *c* winding, D_2, T_2, dc source, as shown in Figure 7.45(i). To commutate phase *a* current, T_3 is gated on. With that, the capacitor voltage is applied across T_1, which reverse biases it. The current will continue to flow through T_3, C_1, D_1, *a* phase, *c* phase, D_2, T_2, dc source. The SCR T_1 is turned off. The current flow reverses the charge across C_1. This is shown in Figure 7.45(ii).

The reversing charge in the capacitor C_1 forward biases D_3; hence, the dc link current is split through both *a* and *b* phases, as shown in Figure 7.45(iii). The voltage across C_1 becomes increasingly negative with respect to D_1, which reverse biases it and cuts off conduction. The dc link current goes through only *b* and *c* phases, as shown in Figure 7.45(iv).

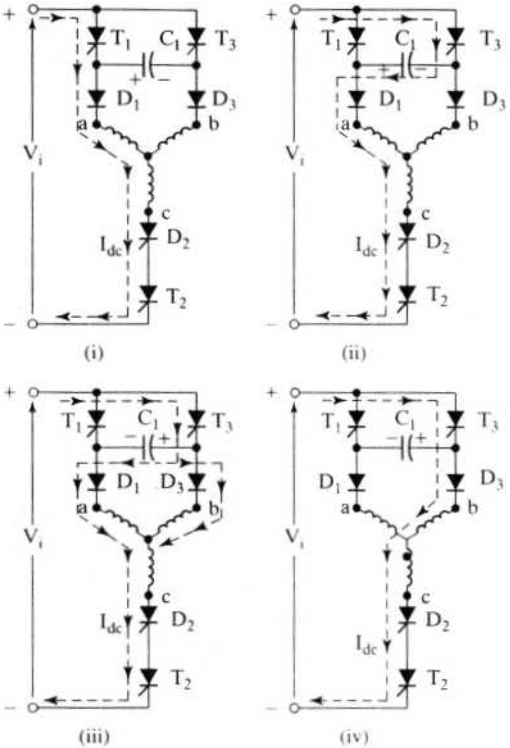


Figure 7.45 Commutation sequence in an autosequentially commutated current-source inverter

The current in phase *a* has been commutated completely. Likewise, gating one SCR will commutate the parallel SCR. The stator phase currents are shown in Figure 7.46 assuming that the stator is Y-connected. For purposes of analysis, it is usual to neglect the rise and fall times of the current and consider only ideal rectangular blocks of current with 120-electrical-degree duration.

The current commutation is slow; hence, converter-grade SCRs with a large turn-off time are sufficient for the ASCI. This reduces the cost of the devices. The transfer of current from one phase to another produces a voltage spike due to the leakage inductances in the motor. Such voltage spikes are undesirable, and they increase the SCR voltage ratings. The commutation capacitors absorb part of this commutation energy and reduce the magnitude of the voltage spikes.

7.5.2.2 Phase-sequence reversal. The phase sequence of the inverter output currents is reversed by changing the sequence of firing the SCRs. The sequence of firing for phase sequence *abc* is $T_1, T_2, \ldots, T_6$; for phase sequence *acb*, it is $T_1, T_6, T_5, T_4, T_3, T_2$. Note that the phase-sequence reversal command is determined by the speed command and actual speed of the induction motor. An abrupt phase-sequence reversal in the induction motor would result in plugging, accompanied by large stator

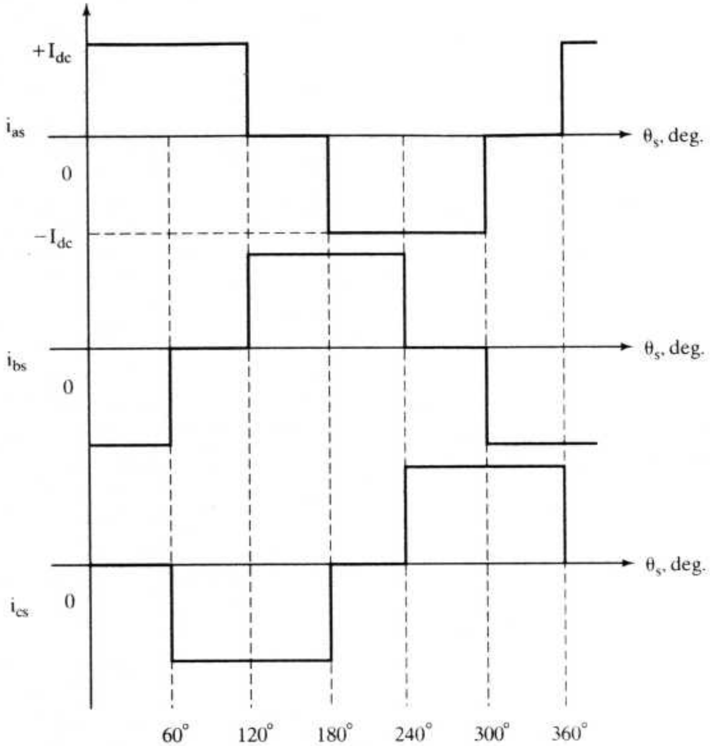


Figure 7.46 Stator currents in a star-connected induction motor fed from a current source

currents. Therefore, a great amount of caution and control is required to initiate such an action.

7.5.2.3 Regeneration. To understand regeneration, it is instructive to go through the motoring operation of the current-source induction motor drive shown in Figure 7.47. In the motoring mode, the electrical power input is positive, and the rectified dc voltage and current are positive. The reflected voltage of the induction motor at the input of the inverter, V_i, is positive, and it opposes the dc input voltage V_r in the dc link. Note that the induction motor drive is in the forward motoring mode with speed and torque in the clockwise direction. In the regeneration mode, the rotor speed remains in the same direction but the torque is reversed. This is achieved by making the slip speed negative. It results in the negative induced emfs in the stator phases of the induction motor, which appear as a negative voltage at the inverter input, as is shown in Figure 7.48. For stable operation, the rectified voltage V_r is reversed, thus enabling the absorption of regenerative energy. The input dc link current is positive, and dc link voltage is made negative, resulting in the input dc link power being negative, implying that the induction machine is sending power to the input ac mains.

The same operation results during the regeneration when the rotor speed is in the counter clockwise direction. Thus, four-quadrant operation is achieved with one

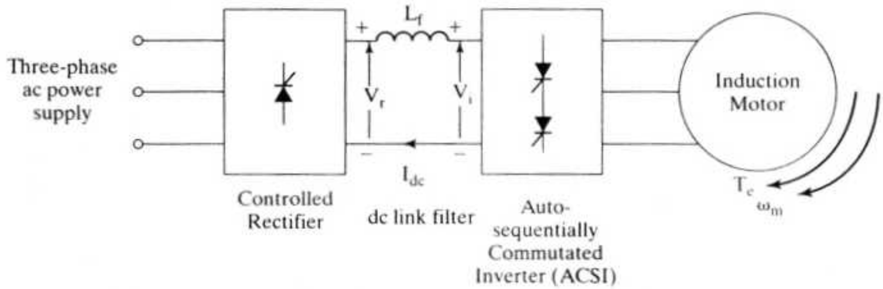


Figure 7.47 Forward motoring of the current-source induction motor drive

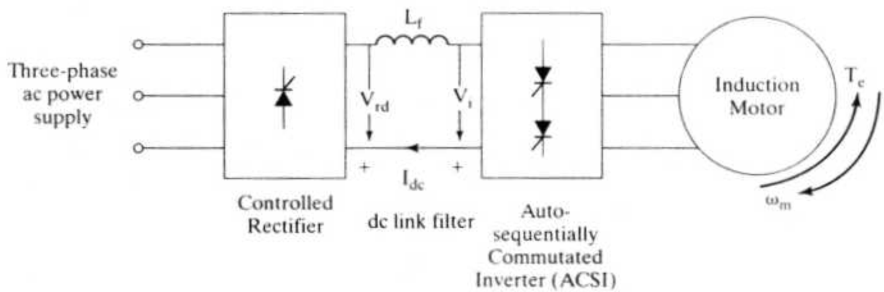


Figure 7.48 Regeneration in the current-source induction motor drive

controlled rectifier, as opposed to a dual controlled rectifier in the case of the voltage-source-inverter induction motor drive.

7.5.2.4 Comparison of converters for ac and dc motor drives. A comparison of converters for dc and ac motor drives is timely to assess the relative cost. The converters are assumed to use SCRs only. The motor drive has to be four-quadrant in operation. The induction motor drives with voltage and current source (ASCI) are compared with the dc motor drives. The comparison is given in Table 7.2. From the table, it is inferred that the ASCI-fed induction motor drive is more economical than its voltage-source counterpart but is more expensive than the dc motor drive. When the cost of the motor is included for comparison, particularly above 100 hp, the ASCI-fed induction motor is likely to be the economical motor drive.

7.5.3 Steady-State Performance

The steady-state performance of the current-source induction motor (hereafter referred to as CSIM) drive can be evaluated approximately by using the fundamental of the input currents or exactly by matching boundary conditions. The approximate solution by using the equivalent circuit of the induction motor lends insight into the influence of the motor parameters on the drive performance. The exact solution using boundary-matching conditions offers no such insight into the drive performance. Both solution methods are developed in this section.

TABLE 7.2 Comparison of SCR converters for four-quadrant dc and voltage- and current-source induction motor drives

			Induction Motor Drives	
Serial Number	Items	DC Drive	Voltage Source	Current Source
1	SCRs	12	36 (full wave)	12
2	Diodes	0	24	6
3	Commutating Reactors	0	6	—
4	Filter Capacitors	0	1	0
5	Commutating Capacitors	0	6 (small)	6 (large)
6	DC Link Filter Inductor	0	0	1 (large)

Equivalent Circuit Approach. The equivalent circuit of the induction motor with constant stator current is shown in Figure 7.49. The rotor and magnetizing currents as a function of stator current are

$$I_r = \frac{jL_m}{\frac{R_r}{s\omega_s} + jL_r} \cdot I_s \tag{7.226}$$

$$I_m = \frac{\frac{R_r}{s\omega_s} + jL_{lr}}{\frac{R_r}{s\omega_s} + jL_r} \cdot I_s \tag{7.227}$$

where the fundamental stator rms current is

$$I_s = \frac{\sqrt{2}\sqrt{3}}{\pi} \cdot I_{dc} = 0.779\, I_{dc} \tag{7.228}$$

and I_{dc} is the dc link current.

The electromagnetic torque is

$$T_e = 3 \cdot \frac{P}{2} \cdot \frac{L_m^2}{\left(\frac{R_r}{s\omega_s}\right)^2 + L_r^2} \cdot \frac{R_r}{s\omega_s} \cdot I_s^2 \tag{7.229}$$

and the maximum torque occurs at

$$s = \frac{R_r}{\omega_s L_r} \tag{7.230}$$

Substituting equation (7.230) into (7.229) yields

$$T_{e(max)} = \frac{3}{2} \cdot \frac{P}{2} \cdot \frac{L_m^2}{\left(\frac{R_r}{s\omega_s}\right)} \cdot I_s^2 = \frac{3}{2} \cdot \frac{P}{2} \cdot \frac{L_m^2}{L_r} \cdot I_s^2 \tag{7.231}$$

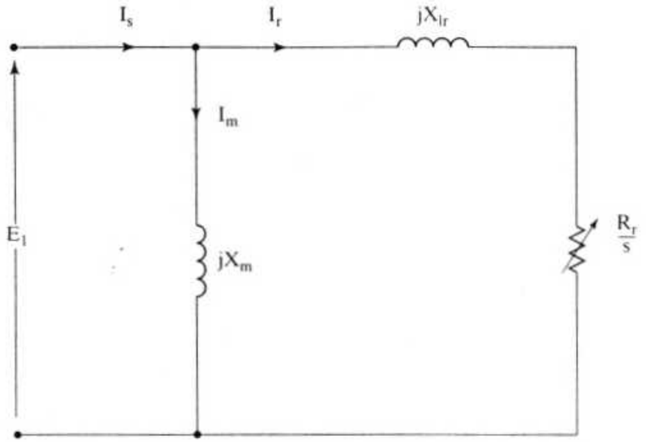


Figure 7.49 Induction-motor equivalent circuit with constant stator current

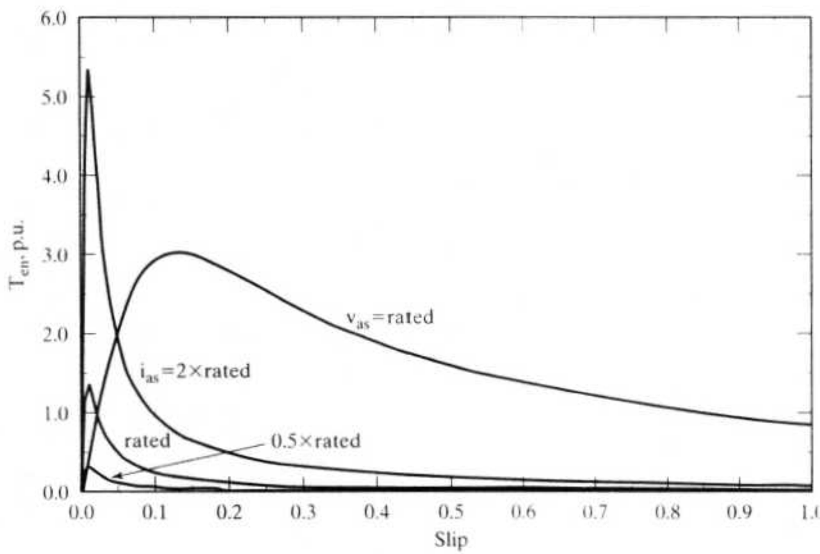


Figure 7.50 Torque–speed characteristics of the current-source induction motor drive for 0.5, 1, and 2 times the rated stator currents and the torque-vs.-slip characteristic at rated stator voltages

The maximum torque, when saturation is considered, becomes much smaller compared to the unsaturated case. From the equivalent circuit, the steady-state torque–speed characteristics are drawn for various values of stator current and shown in Figure 7.50. To contrast them, the torque-vs.-speed characteristics of the voltage-fed induction motor is also drawn in the same figure for nominal stator voltages. This characteristic reflects operation at rated air gap flux linkages if the stator impedance is neglected. The rated air gap-flux-linkages operation is available for the current-source drive only in its statically unstable portion of the characteristic, as is seen from the intersections of the voltage-source and current-source torque–speed characteristics. Points to the left of the intersections indicate operation with higher flux linkages than rated value and, hence, deep saturation. To operate the current-source motor drives at rated flux linkages, correlation between the slip speed and stator current amplitude has to be achieved. Further, these operating points are in the statically unstable region, so stabilization is required in the form of feedback control. That both of these features are unnecessary in the voltage-source inverter-fed induction motor drives is to be taken note of. It further simplifies the control and sensor requirements in the voltage-source drives, thus making them attractive for low-cost, low-performance applications. The steady-state performance without neglecting harmonics is computed by boundary-matching conditions, as given below, and various features of the steady-state performance will be highlighted from these results.

Example 7.10

An induction motor is fed from a current source with base current. The desired operating point is at rated flux that is obtained with a voltage-source feed at base voltage. Find the slip and electromagnetic torque at the desired operating point. The machine details are as follows:

40 hp, 460 V, 4 pole, 3 phase, 60 Hz, Y connected

$$R_r = 0.209\ \Omega;\ L_m = 0.04\ \text{H};\ L_s = 0.0425\ \text{H};\ L_r = 0.043\ \text{H}$$

Ignore the stator resistance and consider that the magnetizing branch is placed ahead of the stator impedance in the equivalent circuit in order to simplify the analytical expressions in this example.

Solution

$$\omega_s = 2\pi f_s = 2\pi * 60 = 376.99\ \text{rad/sec}$$
$$X_m = L_m\omega_s = 15.45\ \Omega$$
$$X_s = L_s\omega_s = 16.02\ \Omega$$
$$X_r = L_r\omega_s = 16.21\ \Omega$$
$$X_{ls} = X_s - X_m;\ X_{lr} = X_r - X_m;$$
$$X_{eq} = X_{ls} + X_{lr} = 1.32\ \Omega$$

Base Values:

$$\text{Base voltage} = 460/\sqrt{3} = 265.6\ \text{V}$$
$$\text{Base current} = P_b/(3V_b) = 40*745.6/(3*265.6) = 37.4\ \text{A}$$

For approximate solution, the induction-motor equivalent circuit shown in Figure 7.49 is chosen as per the problem formulation.

Solution Method The operating point is at the intersection of the voltage and current source torque-vs.-slip characteristics of the induction motor; it satisfies the base current operation from a current source and as well a base flux operation. To find that operating point, the air gap torque expressions for both the current and voltage source are derived and equated to each other. It will yield a solution for slip, and then substitution of the slip in either the current source or voltage source air gap torque expressions will evaluate the operating air gap torque of the machine. The following steps in the solution are given.

For base voltage operation, the air gap torque is

$$T_{ev} = 3\frac{P}{2} I_r^2 \frac{R_r}{s\omega_s} = 3\frac{P}{2} \frac{V_{as}^2}{\left(R_s + \frac{R_r}{s}\right)^2 + X_{eq}^2} \frac{R_r}{s\omega_s}$$

For current-source operation with stator current of base value, the air gap torque is derived from the following:

$$I_s = I_m + I_r$$

$$jX_m I_m = I_r\left(R_s + \frac{R_r}{s} + jX_{eq}\right)$$

and then, finding the rotor current in terms of the stator current, we get

$$I_r = \frac{jX_m}{\left(R_s + \frac{R_r}{s} + jX_{eq} + jX_m\right)} I_s$$

which, when substituted into the torque expression, yields the final relationship:

$$T_{ec} = 3\frac{P}{2} I_r^2 \frac{R_r}{s\omega_s} = 3\frac{P}{2} \frac{X_m^2 I_s^2}{\left(R_s + \frac{R_r}{s}\right)^2 + (X_m + X_{eq})^2} \frac{R_r}{s\omega_s}$$

By equating the torque expressions for current-source and voltage-source operation, the unknown slip is derived. Further, neglecting the stator resistance and keeping the applied voltage as base voltage and the applied current as base current gives the slip as

$$s = \pm R_r \sqrt{\frac{V_b^2 - X_m^2 I_b^2}{(X_m X_{eq} I_b)^2 - (V_b[X_m + X_{eq}])^2}} = 0.0244$$

The air gap torque is obtained by substituting this slip in T_{ec} or T_{ev}:

$$T_e = 128.26\ \text{N}\cdot\text{m}$$

7.5.4 Direct Steady-State Evaluation of Six-Step Current-Source Inverter-Fed Induction Motor (CSIM) Drive System

It is assumed that the stator currents are alternating and rectangular, with 120 electrical degrees of conduction. The rise and fall times of the currents are negligible but would be considered for the computation of the voltage spikes generated by the leakage inductances of the induction motor. The technique for the evaluation of the direct steady-state initial current vector (and hence of the performance of the CSIM drive) is identical to that developed for the six-step voltage-fed induction motor drive described in section 7.4.5.6.

Stator Currents The stator phase currents are shown in Figure 7.46. Viewing them in the synchronous reference frames by using the transformation T^e_{abc} given by the equation (7.41) shows that the quadrature and direct axis stator currents are as follows:

(i) Interval I: $0 < \theta_s < 60°$

$$\begin{aligned} i_{as} &= I_{dc} \\ i_{bs} &= -I_{dc} \\ i_{cs} &= 0 \end{aligned} \tag{7.232}$$

giving the quadrature axis stator current as

$$i^e_{qs} = \frac{2}{3}\left[i_{as}\cos\theta_s + i_{bs}\cos\left(\theta_s - \frac{2\pi}{3}\right) + i_{cs}\cos\left(\theta_s + \frac{2\pi}{3}\right)\right] \tag{7.233}$$

yielding for the first interval the quadrature axis current, which is indicated by an additional subscript of I:

$$i^e_{qsI} = aI_{dc}\cos\left(\theta_s + \frac{\pi}{6}\right) \tag{7.234}$$

where

$$a = \frac{2}{\sqrt{3}} \tag{7.235}$$

Similarly, for the direct axis stator current in the synchronous reference frames, we obtain

$$i^e_{dsI} = a\mathbf{I}_{dc}\sin\left(\theta_s + \frac{\pi}{6}\right) \tag{7.236}$$

(ii) interval II: $\frac{\pi}{3} < \theta_s < \frac{2\pi}{3}$

The quadrature and direct axis stator currents by transformation then are given with an additional subscript of II:

$$\begin{aligned} i^e_{qsII} &= a\mathbf{I}_{dc}\cos\left(\theta_s - \frac{\pi}{6}\right) \\ i^e_{dsII} &= aI_{dc}\sin\left(\theta_s - \frac{\pi}{6}\right) \end{aligned} \tag{7.237}$$

and they could be written to resemble the first interval quadrature and direct axis stator currents as

$$\begin{aligned} i^e_{qsII} &= aI_{dc}\cos\left(\left\{\theta_s + \frac{\pi}{6}\right\} - \frac{\pi}{3}\right) \\ i^e_{dsII} &= aI_{dc}\sin\left(\left\{\theta_s + \frac{\pi}{6}\right\} - \frac{\pi}{3}\right) \end{aligned} \tag{7.238}$$

In this form, it is evident that the q and d axes currents are periodic; hence, the currents in the second interval can be obtained from the currents in the first interval. Thus knowledge of the currents in one 60-degree interval is sufficient to reconstruct the current for one cycle. Expanding the right-hand side of the second-interval q and d axes currents results in the terms of the first interval q and d currents as

$$i^e_{qsII} = aI_{dc}\left[\cos\left(\theta_s + \frac{\pi}{6}\right)\cos\frac{\pi}{3} + \sin\left(\theta_s + \frac{\pi}{6}\right)\sin\frac{\pi}{3}\right] = \frac{1}{2}\left\{aI_{dc}\cos\left(\theta_s + \frac{\pi}{6}\right)\right\}$$

$$+ \frac{\sqrt{3}}{2}\left\{aI_{dc}\sin\left(\theta_s + \frac{\pi}{6}\right)\right\} = \frac{1}{2}i^e_{qsI} + \frac{\sqrt{3}}{2}i^e_{dsI} \tag{7.239}$$

Similarly, the direct axis stator current in the second interval is derived in terms of the first interval q and d axes currents as.

$$i^e_{dsII} = -\frac{\sqrt{3}}{2}i^e_{qsI} + \frac{1}{2}i^e_{dsI} \tag{7.240}$$

Combining the above two equations in matrix form and generalizing between any two consecutive intervals of 60 degrees yields

$$\begin{bmatrix} i^e_{qs}\left(\theta_s + \frac{\pi}{3}\right) \\ i^e_{ds}\left(\theta_s + \frac{\pi}{3}\right) \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}\begin{bmatrix} i^e_{qs}(\theta_s) \\ i^e_{ds}(\theta_s) \end{bmatrix} \tag{7.241}$$

Now, because they are continuous functions, they are differentiable, thus providing the following relationship in state-space form among themselves:

$$\dot{X}_1 = S_1X_1 \tag{7.242}$$

where the matrix S_1 and vector X_1 are given by

$$X_1 = [i^e_{qs} \quad i^e_{ds}]^T \tag{7.243}$$

$$S_1 = \begin{bmatrix} 0 & -\omega_s \\ \omega_s & 0 \end{bmatrix} \tag{7.244}$$

Machine Equations With stator currents as the inputs, the performance evaluation requires only the solution of rotor currents from the rotor voltage equations. Note that the stator currents are discontinuous in real time and rotor flux linkages are continuous. The rotor equations with rotor flux linkages as variables are used in the solution of the rotor variables and hence in the computation of the electromagnetic torque. The rotor equations in rotor flux linkages are derived as follows. The rotor equations in the synchronous reference frames are

$$R_ri^e_{qr} + p(L_ri^e_{qr} + L_mi^e_{qs}) + \omega_{sl}(L_ri^e_{dr} + L_mi^e_{ds}) = 0 \tag{7.245}$$

$$R_ri^e_{dr} + p(L_ri^e_{dr} + L_mi^e_{ds}) - \omega_{sl}(L_ri^e_{qr} + L_mi^e_{qs}) = 0 \tag{7.246}$$

Defining the rotor flux linkages in the q and d axes as

$$\lambda_{qr}^e = L_r i_{qr}^e + L_m i_{qs}^e \tag{7.247}$$

$$\lambda_{dr}^e = L_r i_{dr}^e + L_m i_{ds}^e \tag{7.248}$$

and substituting these into the rotor equations yields the equations in rotor flux linkages as

$$p\lambda_{qr}^e + \omega_{sl}\lambda_{dr}^e + R_r i_{qr}^e = 0 \tag{7.249}$$

$$p\lambda_{dr}^e - \omega_{sl}\lambda_{qr}^e + R_r i_{dr}^e = 0 \tag{7.250}$$

and the rotor currents are obtained from equations (7.247) and (7.248) as

$$i_{qr}^e = \frac{1}{L_r}(\lambda_{qr}^e - L_m i_{qs}^e) \tag{7.251}$$

$$i_{dr}^e = \frac{1}{L_r}(\lambda_{dr}^e - L_m i_{ds}^e) \tag{7.252}$$

Substitution of these rotor currents into the rotor flux-linkages equations results in the following equations in rotor flux linkages:

$$p\lambda_{qr}^e + \frac{R_r}{L_r}\lambda_{qr}^e + \omega_{sl}\lambda_{dr}^e - R_r\frac{L_m}{L_r}i_{qs}^e = 0 \tag{7.253}$$

$$-\omega_{sl}\lambda_{qr}^e + p\lambda_{dr}^e + \frac{R_r}{L_r}\lambda_{dr}^e - R_r\frac{L_m}{L_r}i_{ds}^e = 0 \tag{7.254}$$

Combining these equations with the state equations of the stator currents casts the system equations in a compact state-space form:

$$\dot{X} = AX \tag{7.255}$$

where the state vector X and state matrix A are given by

$$X = [\lambda_{qr}^e \quad \lambda_{dr}^e \quad i_{qs}^e \quad i_{ds}^e]^t \tag{7.256}$$

$$A = \begin{bmatrix} -\dfrac{R_r}{L_r} & -\omega_{sl} & \dfrac{R_r L_m}{L_r} & 0 \\ \omega_{sl} & -\dfrac{R_r}{L_r} & 0 & \dfrac{R_r L_m}{L_r} \\ 0 & 0 & 0 & -\omega_s \\ 0 & 0 & \omega_s & 0 \end{bmatrix} \tag{7.257}$$

The solution of the state-space equation is given by

$$X(t) = e^{At}X(0) \tag{7.258}$$

The symmetry of the rotor flux linkages and the stator currents for every 60 electrical degrees shows that the state vector at 60 degrees could be equated to the boundary-matching condition times the initial vector, as follows:

$$X\left(t = \frac{\pi}{3\omega_s}\right) = e^{A\frac{\pi}{3\omega_s}}X(0) = VX(0) \tag{7.259}$$

where

$$V = \begin{bmatrix} I & 0 \\ 0 & S_1 \end{bmatrix} \tag{7.260}$$

where I is an identity matrix of 2×2 dimensions and S_1 has been defined above. Rearranging the above equation as a function of $X(0)$ yields

$$\left[V - e^{A\frac{\pi}{3\omega_s}}\right]X(0) = 0 \tag{7.261}$$

We can rewrite this equation in a compact way as

$$WX(0) = 0 \tag{7.262}$$

where

$$W = \left[V - e^{A\frac{\pi}{3\omega_s}}\right] = \begin{bmatrix} w_1 & w_2 \\ w_3 & w_4 \end{bmatrix} \tag{7.263}$$

The submatrices w_1, w_2, w_3, and w_4 are of 2×2 dimensions, from which the initial steady-state rotor flux-linkages vector is obtained as

$$x_1(0) = -w_1^{-1}w_2x_2(0) \tag{7.264}$$

where the vectors x_1 and x_2 are defined as

$$x_1 = [\lambda_{qr}^e \quad \lambda_{dr}^e]^t \tag{7.265}$$

$$x_2 = [i_{qs}^e \quad i_{ds}^e]^t \tag{7.266}$$

Once the initial steady-state vector is found, compute the exponential for a small interval of time corresponding to 2 or 3 electrical degrees from the exponential series, considering 7 to 10 terms (depending on the accuracy of the solution desired), and store this matrix for repetitive computation of the state variables via the following recursive relationship:

$$X(nT) = e^{AT}X(\{n - 1\}T) \text{ for } n = 0, 1, \ldots, k \tag{7.267}$$

where T is the small sampling time of interest and k is related to it by

$$k = \frac{\pi}{3\omega_s T} \tag{7.268}$$

Note that by storing n of the X vectors and using the boundary-matching condition matrix V, the state vectors for the rest of the five of the 60-degree intervals are generated, and from them the electromagnetic torque is found to be

$$T_e(t) = \frac{3}{2}\frac{P}{2}\frac{L_m}{L_r}(i_{qs}^e\lambda_{dr}^e - i_{ds}^e\lambda_{qr}^e) \tag{7.269}$$